

EXCEL 2021

For Beginners



A Step by Step Practical Guide on How to Use and
Master Microsoft Excel's Features, Functions,
and Formulas

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INTRODUCTION

Microsoft Excel is the world's most popular spreadsheet program with millions of users. Excel and other spreadsheet tools are excellent for data processing, analysis, and visualization because they can sort, filter, format, and display data in one place. Is it possible to calculate an average score automatically? How do I collect contact information for a trip? Or can you help students with lab research reports? Then Excel is an essential tool for you and your students!

Data is everywhere, and our job is to understand it. Interact in real-time with colleagues and students using the free online version of Excel or the extended desktop app. Excel also has natural language, data analysis, and practical tools to help you quickly understand your data.

Official release date?

Microsoft 365's Corporate Vice President announced that Office 2021 would be available to users later this year. This means that it may be become available anytime between now and the end of 2021. However, prior releases have given us a good guess on when to expect it. Office 2019 and 2016 were both released in September of their respective years, so a similar timeline appears to be in the works.

Microsoft Office 2021 new features

Microsoft has released the first details about the upcoming edition of Office, including an updated user interface, new tools that allow users to modify data more efficiently in Excel, and other improvements.

The release of this program has been discussed in depth. This program has been separated from Microsoft's cloud-based productivity suite, Microsoft 365. The core applications are the same in both; however, Office is available as a one-time purchase while Microsoft 365 is available as a subscription-based service.

Office 2021 will improve the performance of the suite's applications. There are also new accessibility features like Dark Word Mode that Microsoft hasn't revealed yet. When customers use Office on their laptops, this option can help save energy and reduce screen brightness. Microsoft provided little details about the enhancements to other Office products that this version will provide. Still, it did tell The Verge that Excel will get two new features that will make working with spreadsheets easier and faster.

The first is known as a dynamic array, and it enhances Excel's data analysis capability. Compressing a single cell in the spreadsheet may cause formatting concerns for complex business calculations that provide several results. The dynamic array automatically groups the results into a single cell.

Excel also comes with a feature called XLOOKUP, which allows you to search for data in a single row of the worksheet instead of displaying the entire spreadsheet, which is helpful in complex documents with many similar items.

How is Microsoft Office 2021 different from Microsoft 365?

Outside of the pricing arrangement, the most important distinction between both programs is the intended use. Employees who don't have regular Internet access and can't benefit from Microsoft 365's cloud-based capabilities typically have Office installed on their devices. For example, computers needed to control factory manufacturing facilities can operate in an isolated network for security reasons.

Office 2021 will be available in multiple versions; the consumer and small business versions will be released at the same price as the current version of Office, while large businesses will pay a 10% premium. Another change that may affect business users is that Microsoft will only support Office 2021 for five years rather than the previous seven years. Customers will need to upgrade ahead of time to receive software updates.

There are several ways in which you can get Microsoft Excel. You can head to the closest computer store and ask for the software version. Or, you can download it from Microsoft's website, but you'll need to buy a license key to use it.

CHAPTER ONE- GETTING STARTED

How to Open Microsoft Excel

You can launch Excel just like any other Windows program. The following steps will work with Windows XP, Vista, and 7.

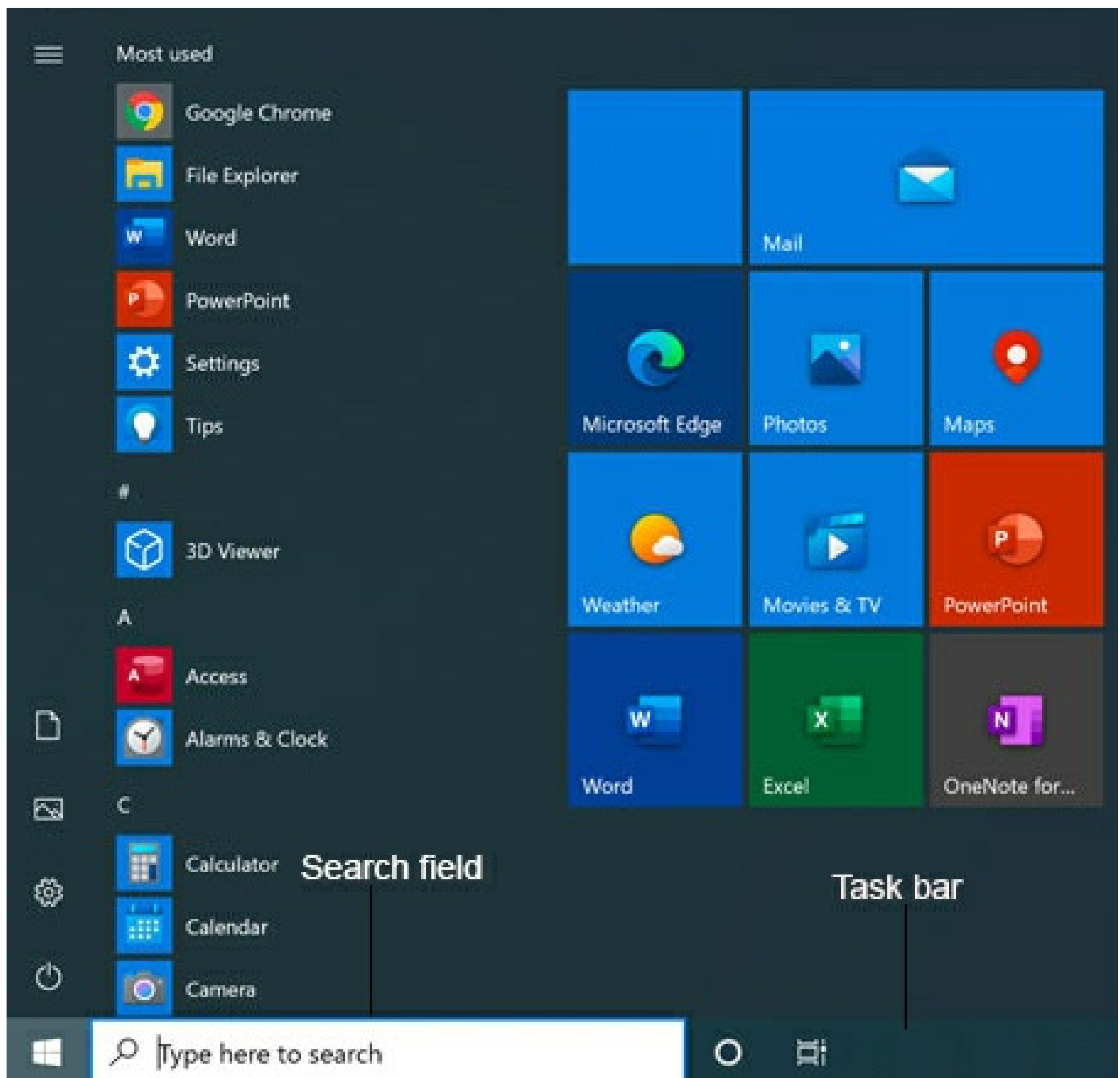
- Press the Windows button to display your Start Menu, go to “ All Programs,” and then click on “Microsoft Excel” to launch it

For Windows 8.1

- Press the Windows button to display your Start Menu, search for Excel, and then click on “Microsoft Excel” when it appears on the list

For Windows 10

- Press the Windows key on your keyboard, open the list of apps on your device and then scroll down to the “E” section.
- Click on Excel to launch it.
- Or, press the Windows key and then use the search field to locate it quickly.
- To make it easier to launch the app, you can also pin an Excel icon to the Windows 10 taskbar. Head to the Excel icon, right-click on it, and then tap on “Pin to Taskbar.”

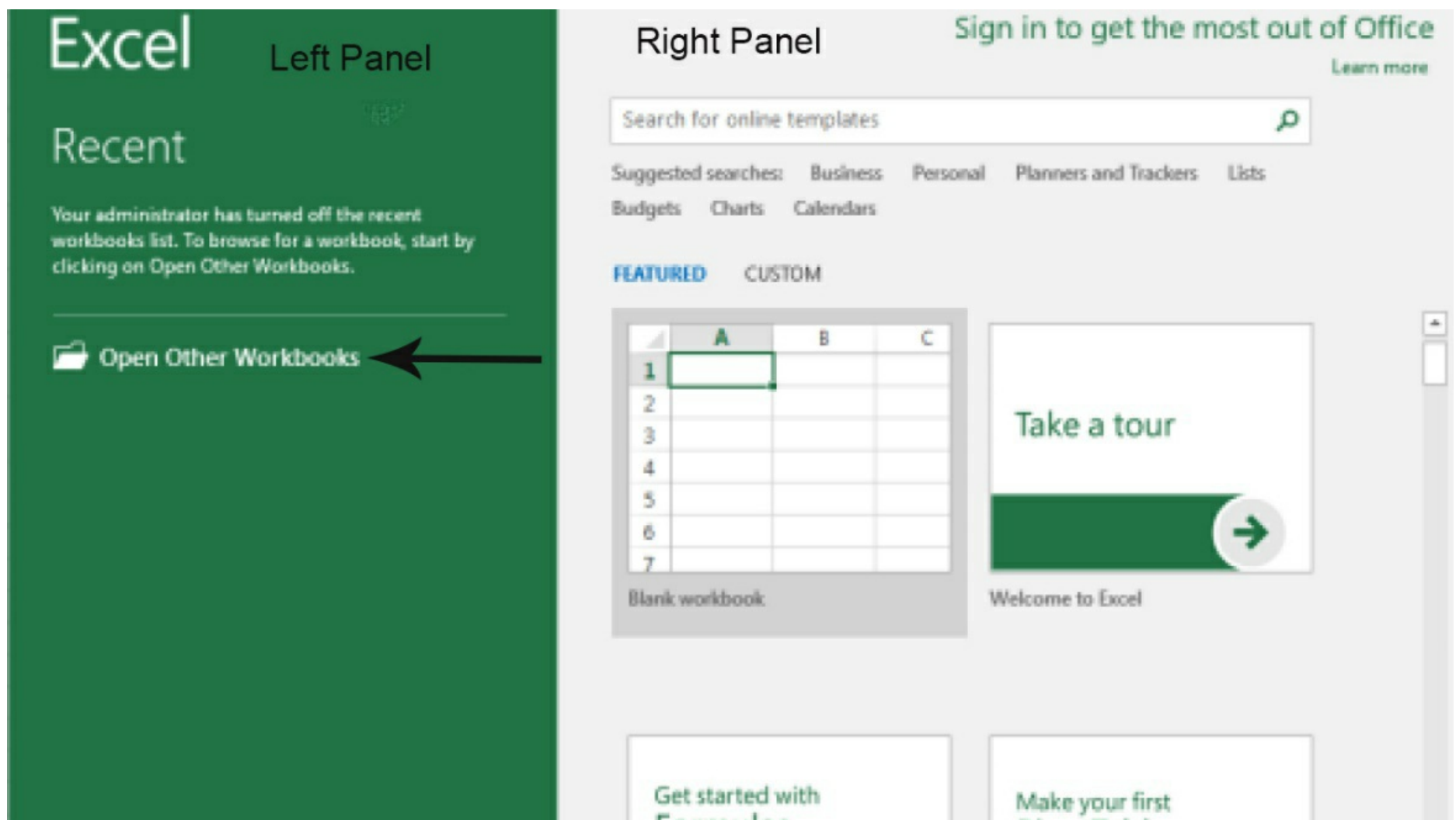


Exiting Excel 2021

- If you are done with your work, make sure it is saved before shutting down the program:
- Press Alt+F4 on your keyboard to close it.
- Or, head to the Excel icon on the taskbar, right-click on the icon and then tap on “Close Window.”
- If you try to close the program without saving your work, the program will alert you telling you to save your changes. Before closing the program, click on the Save button or click on “Don’t Save” to exit the program without saving your last changes .

Excel's Start Screen

When you first run this program, it will show you the default startup screen, which looks like the one below.

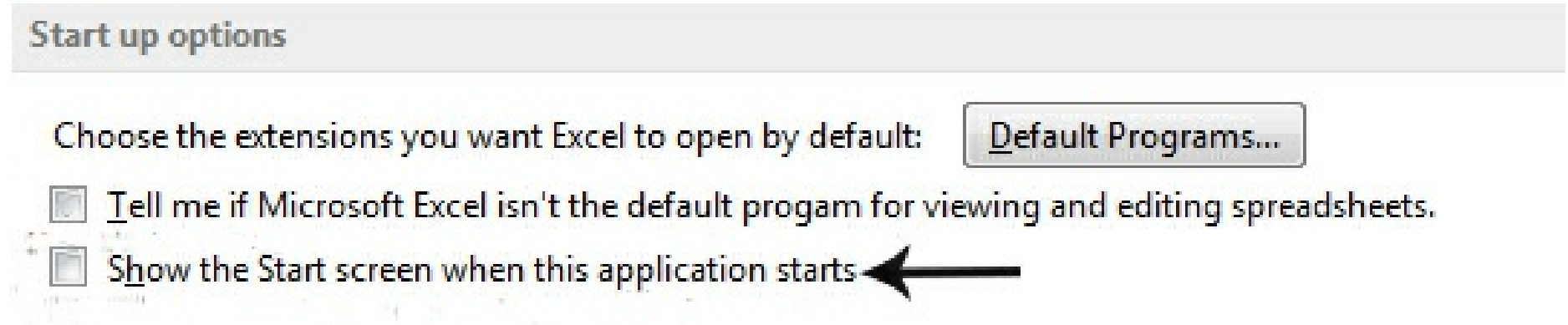


- The left panel contains a list of all your recently opened files.
- Clicking on "Open Other Workbooks" will send you to the Backstage view where you can open any of your existing excel document.
- On the right panel, you'll see a list of standard templates available in Excel. If none of them are what you're looking for, simply type in the keyword you're looking for.
- Most of the popular phrases can also be found below the bar. Excel will begin searching for templates in your selected category.
- Simply use the shortcut "Ctrl + N" to create a new document or Click on "Blank Workbook."

Turn off the Start screen

If the start screen gets in your way and you'd like to skip it, here's how:

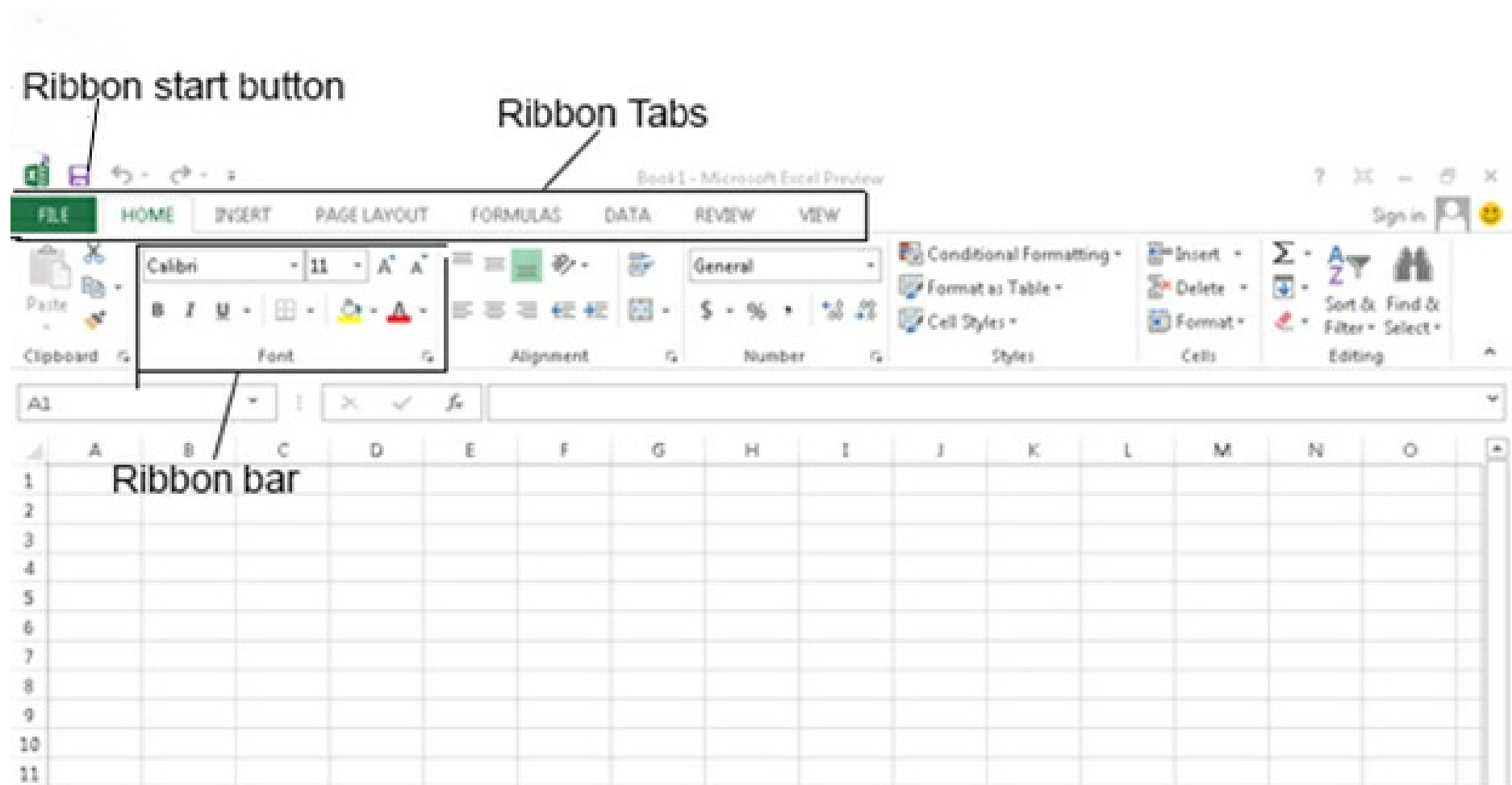
- Open Excel. Go to File, go to Options, and then uncheck "Show the Start screen when this application starts." That's it!
- Repeat the same steps to turn on the Start screen.



Excel's Ribbon User Interface

The Ribbon gives shortcuts to commands. A command is an action performed by the user. Creating a new document, printing a document, and so on are examples of commands. Let's take a closer look at each component of the Ribbon.

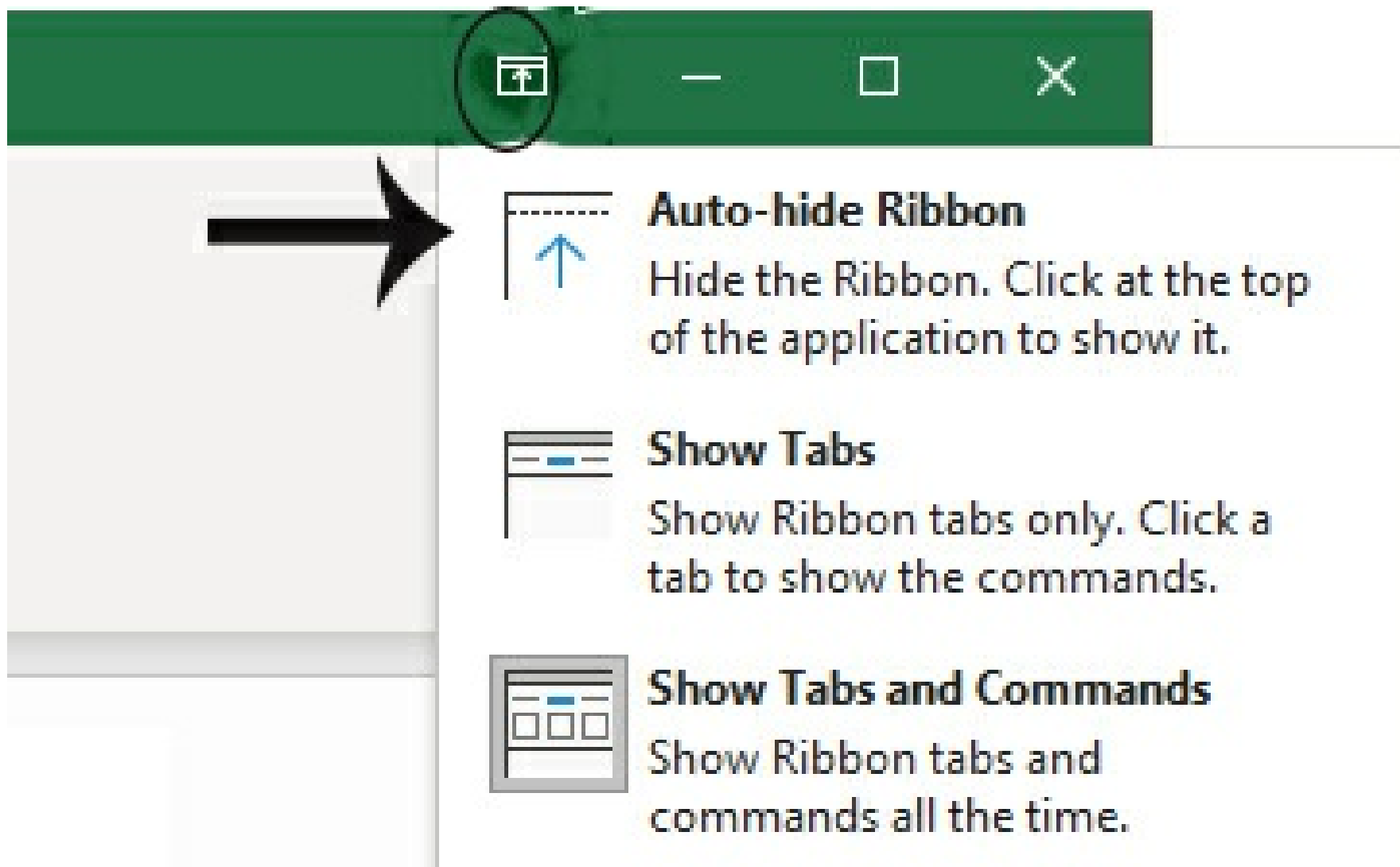
- Start button: Here, you can create a new document, save your work, print, and access other commands.
- Tabs: Tabs are used to organize similar commands. Basic instructions such as formatting data to make it easier to browse, sort, and identity specific data in the table are performed on the "Home" tab.
- Bars: Bars are used to group similar commands. The Font ribbon bar, for example, is used to change your font style, font size, color font, bold font, and more.



How to change the Ribbon Display Options

Although the Ribbon is designed to respond to your current work, you can minimize it if you feel it takes up too much screen space.

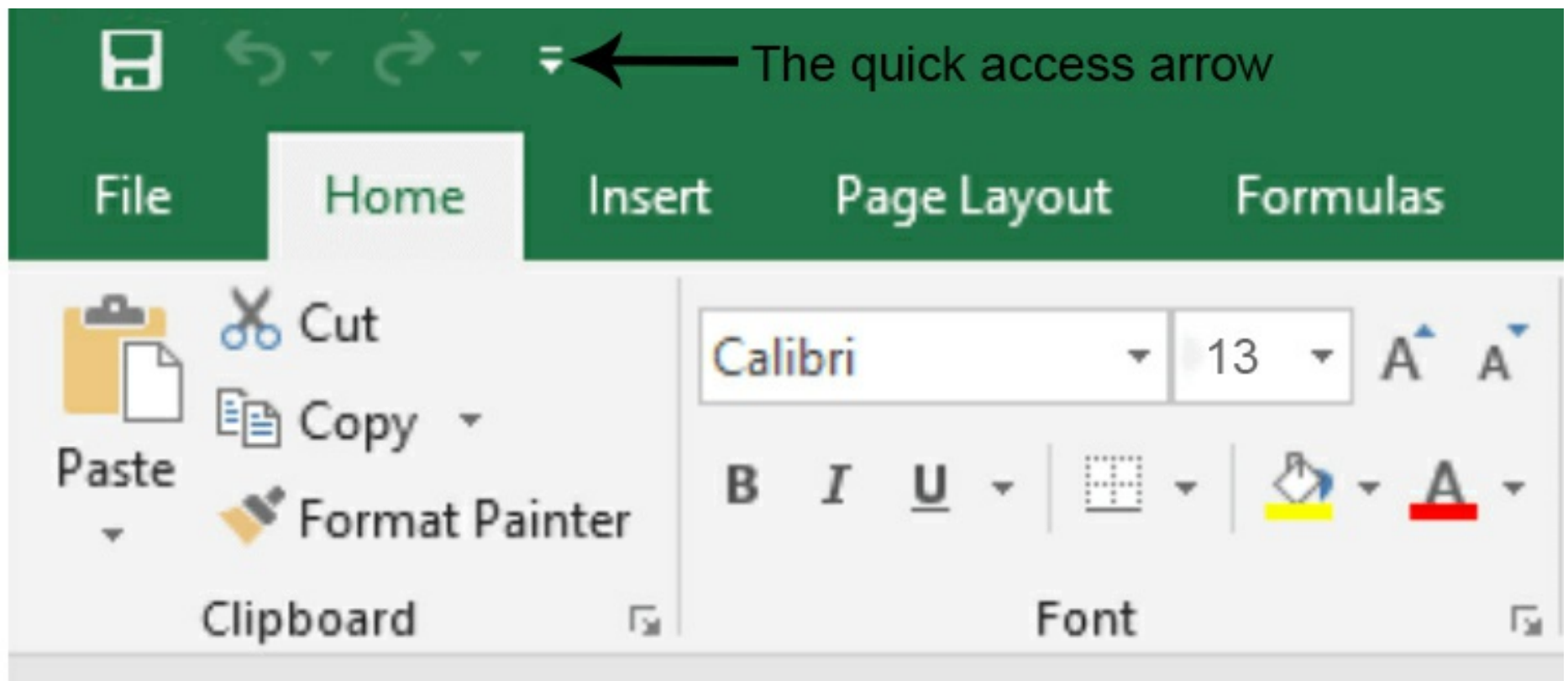
- Click on the "Ribbon Display Options" arrow and then select one of the options from the menu.
 - Auto-hide Ribbon: This will hide the Ribbon and turn your workbook to full-screen.
 - Show Tabs: When this option is enabled, all command groups will be hidden when not in use, although tabs remain visible. Click on a tab to bring up the Ribbon.
 - Show Tabs and Commands: This is the default excel option, and it maximizes the Ribbon, showing all your tabs and commands.



The Quick Access Toolbar

With this option, you'll have access to general commands regardless of the tab you're on. Default commands such as Save, Undo, and Repeat. Other commands can be added as needed. Here's how to do that:

- Click on the drop-down arrow to the right of the Quick Access Toolbar to open a list of commands.



- Select a command from the list or click on “More Commands” to view other available options. That’s it!

Customize Quick Access Toolbar

New

Open

✓ Save

Email

Quick Print

Print Preview and Print

Spelling

✓ Undo

✓ Redo

Sort Ascending

Sort Descending

Touch/Mouse Mode

More Commands... 

Show Below the Ribbon

Understanding the Worksheet

A worksheet is a set of columns and rows. Cells are formed at the intersection of a column and row. Data is recorded in cells. Letters are used to mark columns, while numbers are used to label rows.

The term "workbook" refers to a collection of worksheets. The Excel workbook comes with three cells by default. As needed, you can delete or add new worksheets. Change the worksheet's name from Sheet1 to something more practical, such as Test score or Shopping Budget.

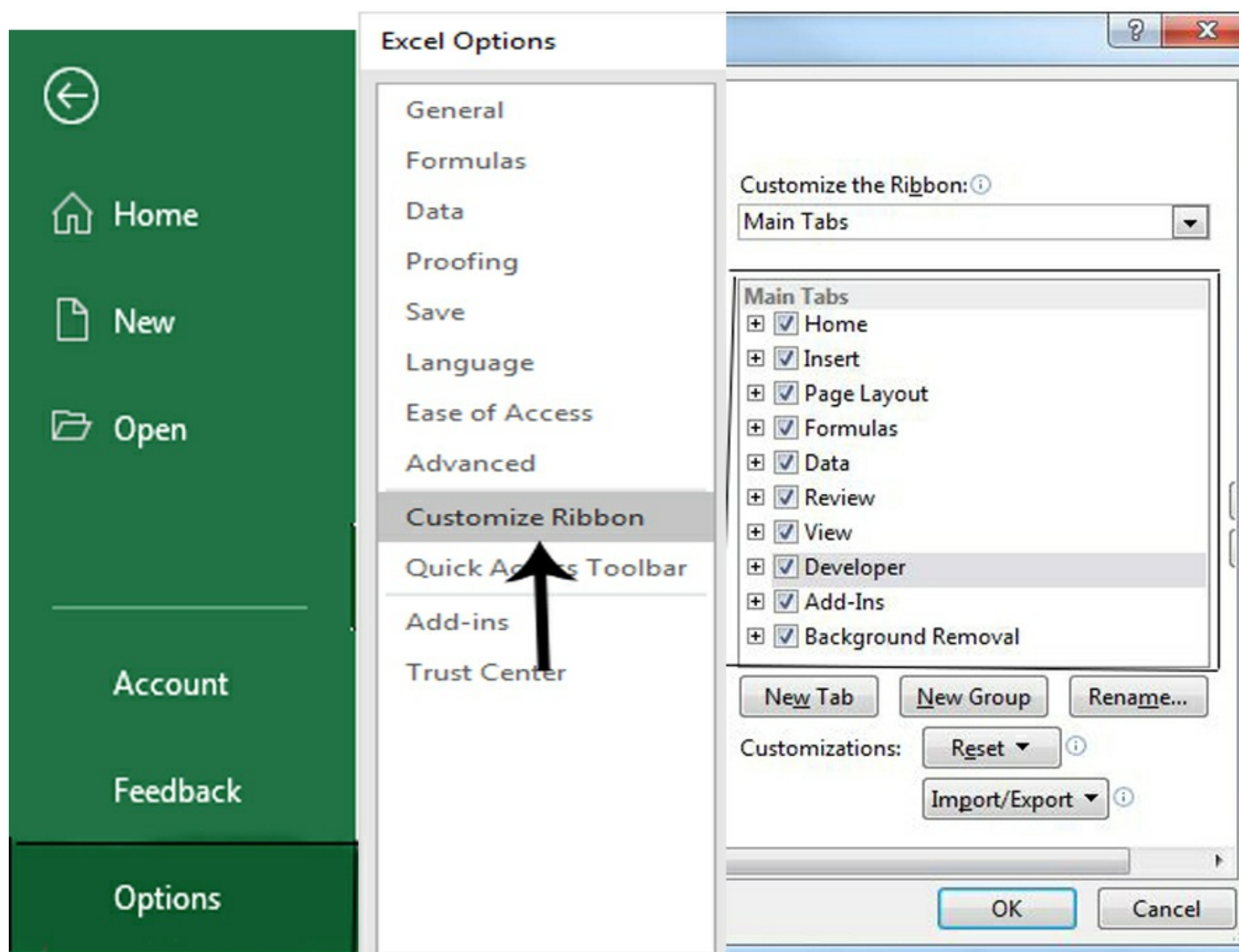
How to customize your Excel environment

I prefer my Excel theme to look dark. But you can customize your theme color to blue or any other color. Let's get on with giving your workbook a unique look.

1. How to customize the ribbon

Now, here's how to delete or add useful tabs to your ribbon:

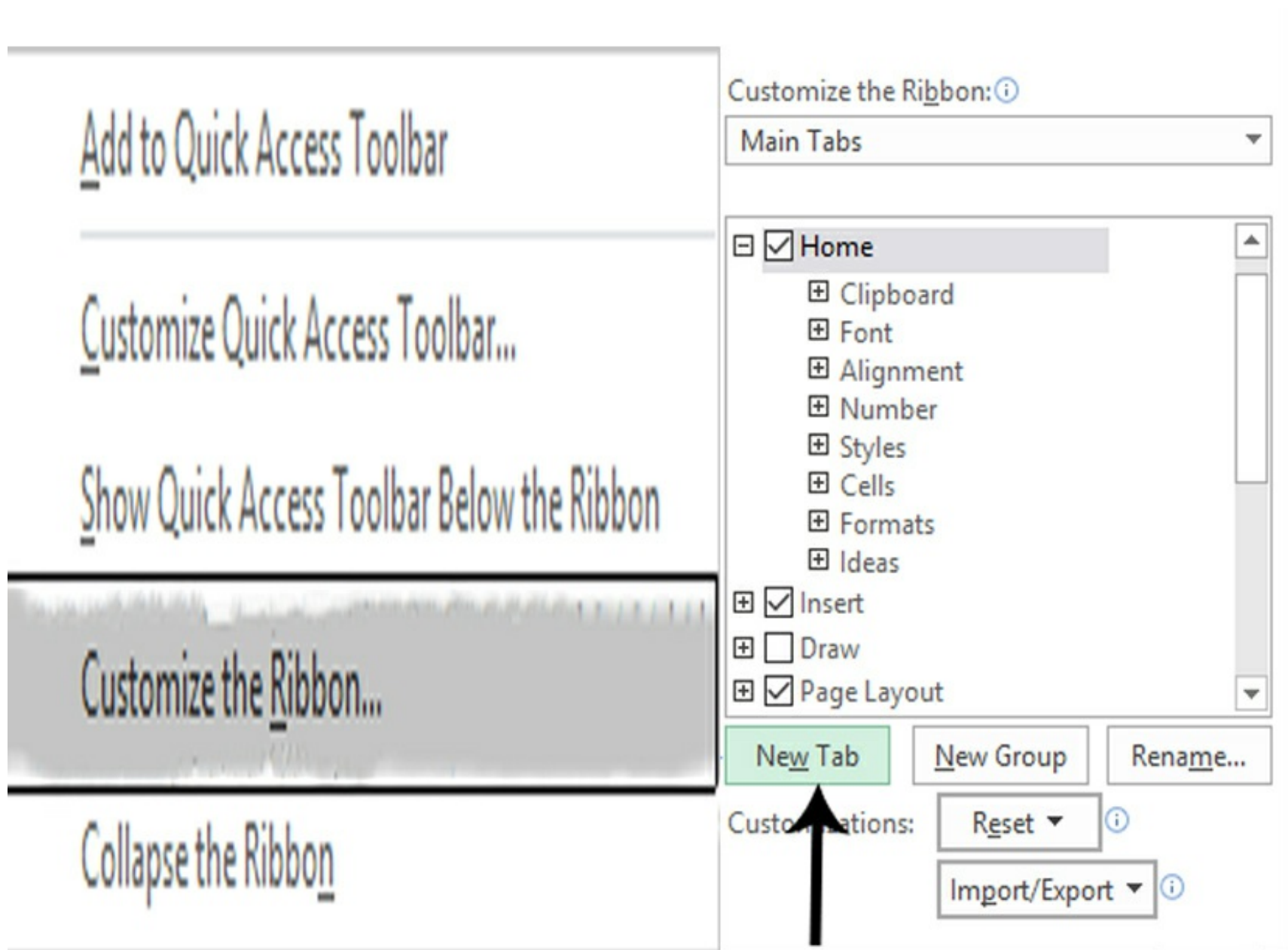
- First, go to the ribbon start button or the Save icon, click on Options, and click on the "Customize Ribbon" option.
- Next, go to the Main tab options, unmark any tab you don't want to use, and click "OK" to save. Your ribbon should look different now.



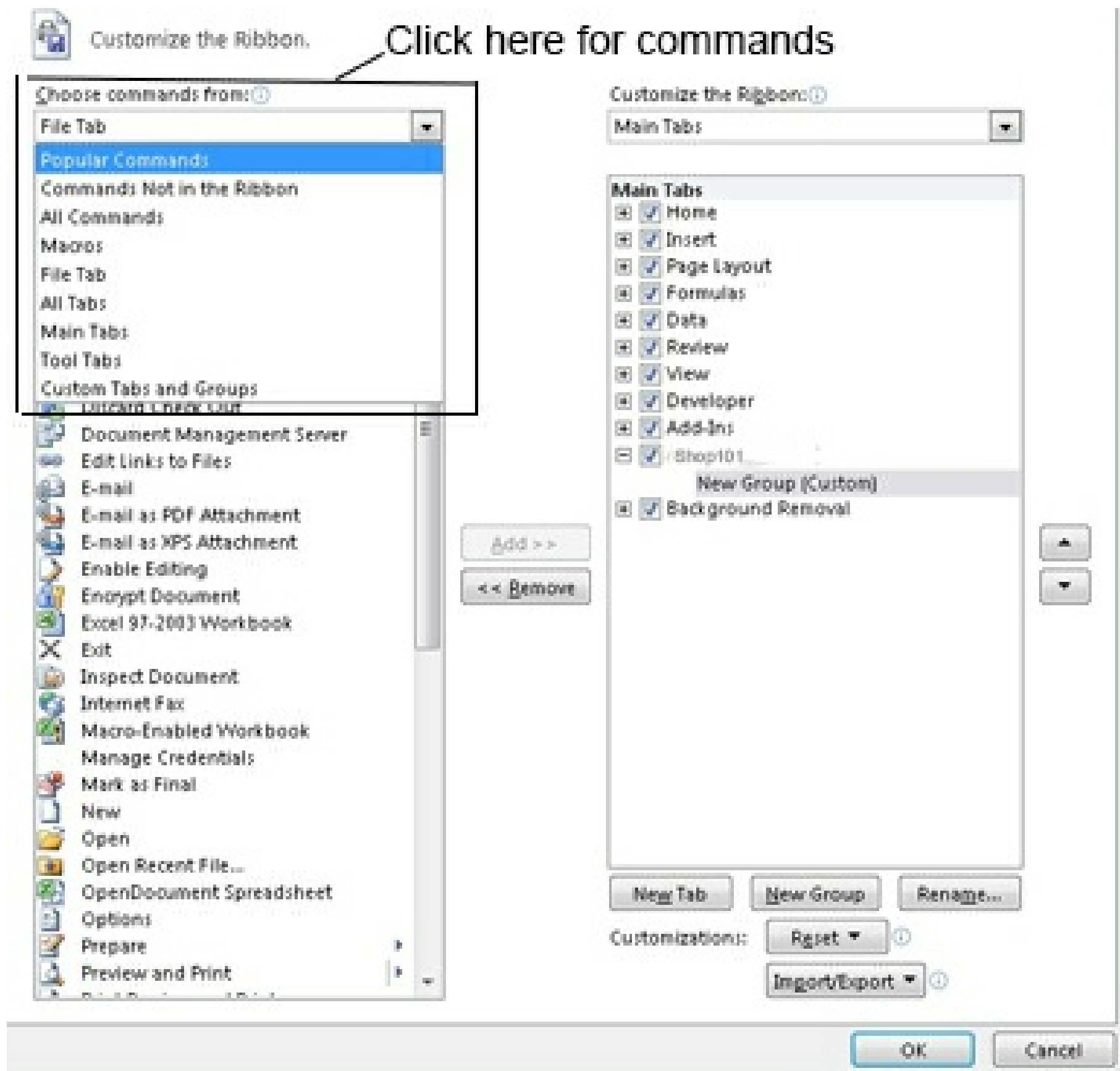
2. How to add custom tabs

Now, let's say we need a new tab named "Shop101" on the ribbon. Here's how to create the tab and assign a command to it:

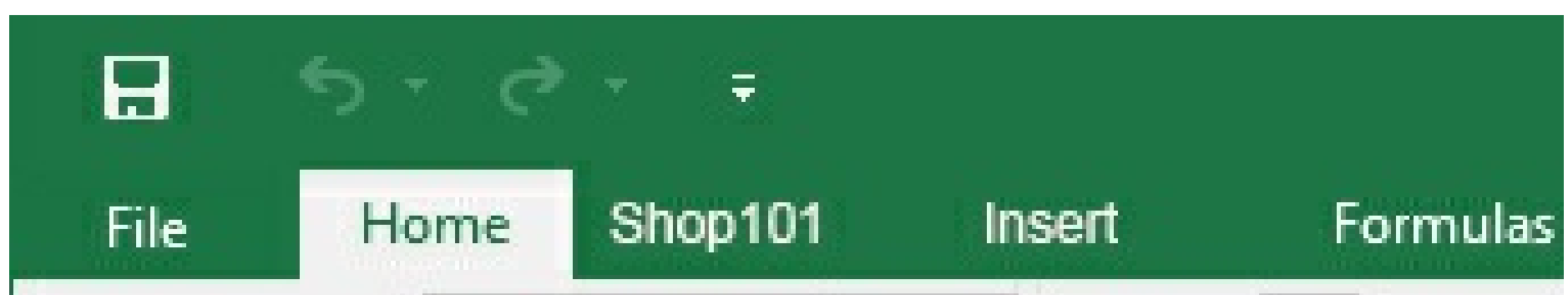
- Right-click on the Ribbon/Save button, click on "Customize the Ribbon," head to the "Main Tabs" option, and then click on "New Tab"



- Next, select your new tab, and then using the Rename button, rename it to Shop101
- Select the New Group (Custom) under Shop101, click on the Rename button and then change the name to "My Commands"
- Now, check the middle panel for the commands you want to add.



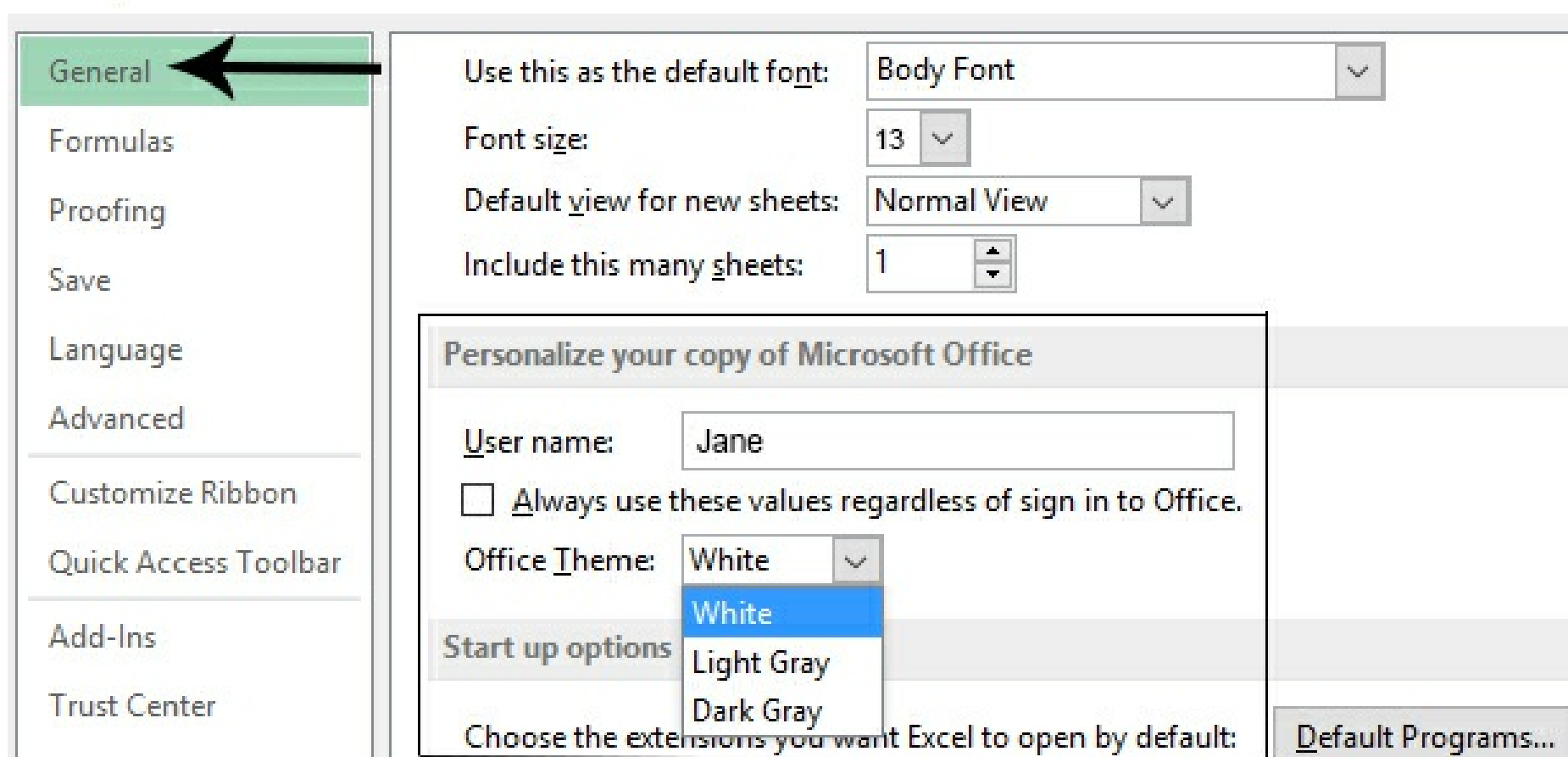
- Select the “All chart types” command, click on the Add button, and then click OK to save. That’s it. Your ribbon should look like this”



3. Setting the color theme

- Click on File or click on the Ribbon button and then go to Options.
- Next, look for the “Office Theme” option, click on its drop-down arrow, and then select your preferred color.
- Click the OK button to save.

Excel Options

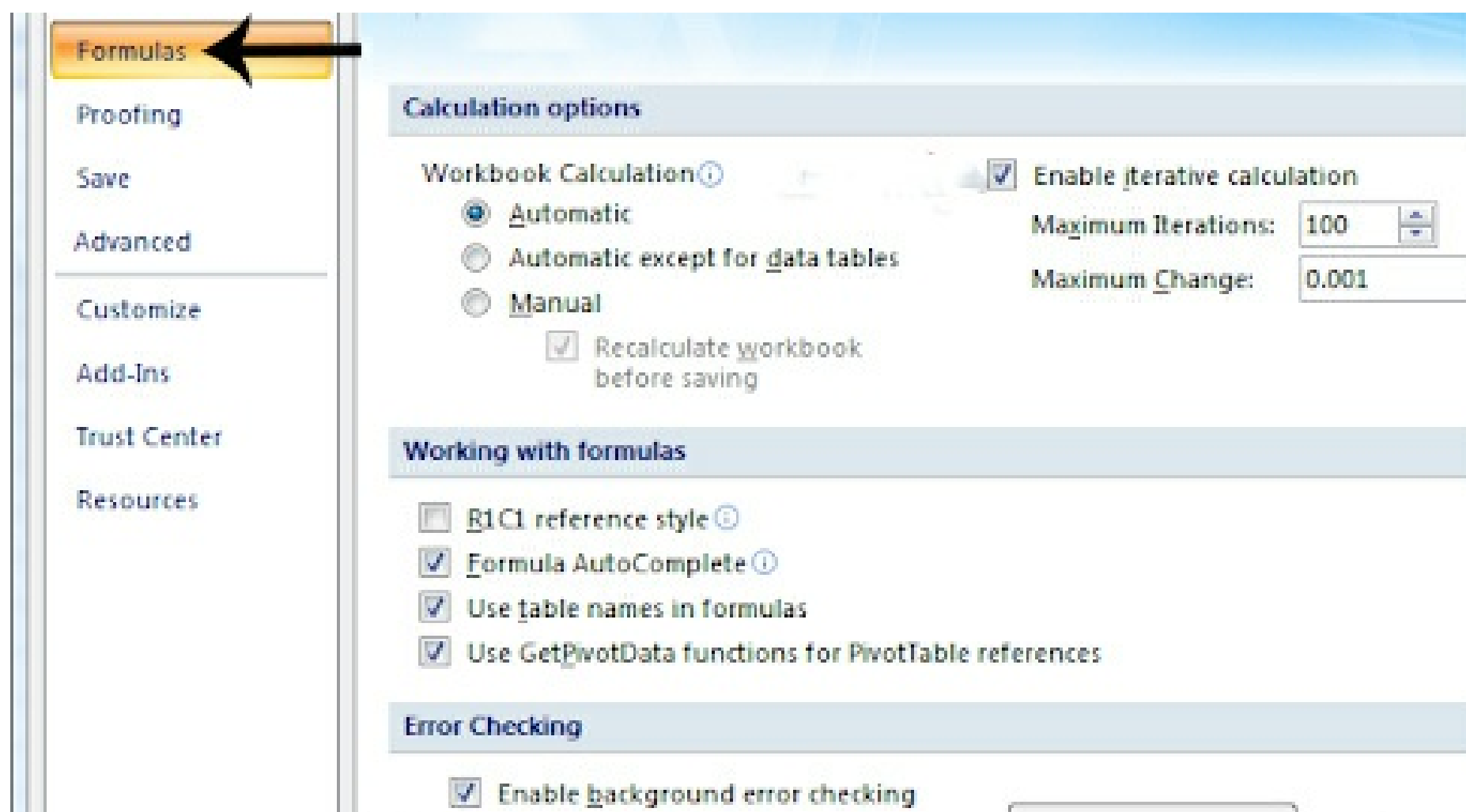


4. Settings for formulas

This option allows you to change the behavior of Excel when dealing with formulas. It can configure settings such as autocomplete when inputting formulas, changing the cell referencing style, using numbers for both columns and rows, etc.

To make an option active:

- Check the box next to it.
- Remove the checkmark from the checkbox to disable an option.
- Avoid using formulas that contain many variables.
- Break them down into smaller and manageable results.



5. Proofing settings

This option alters the text that has been typed into Excel. It allows you to customize things like the dictionary language used when checking for grammatical errors, dictionary suggestions, etc.

General

Formulas

Data

Proofing

Save

Language

Ease of Access

Advanced

Customize Ribbon

Quick Access Toolbar

Add-ins

Trust Center

abc

Change how Excel corrects and formats your text.

AutoCorrect options

Change how Excel corrects and formats text as you type: AutoCorrect Options...

When correcting spelling in Microsoft Office programs

☒ Ignore words in UPPERCASE

☒ Ignore words that contain numbers

☒ Ignore Internet and file addresses

☒ Flag repeated words

☐ Enforce accented uppercase in French

☐ Suggest from main dictionary only

Custom Dictionaries...

French modes: Traditional and new spellings

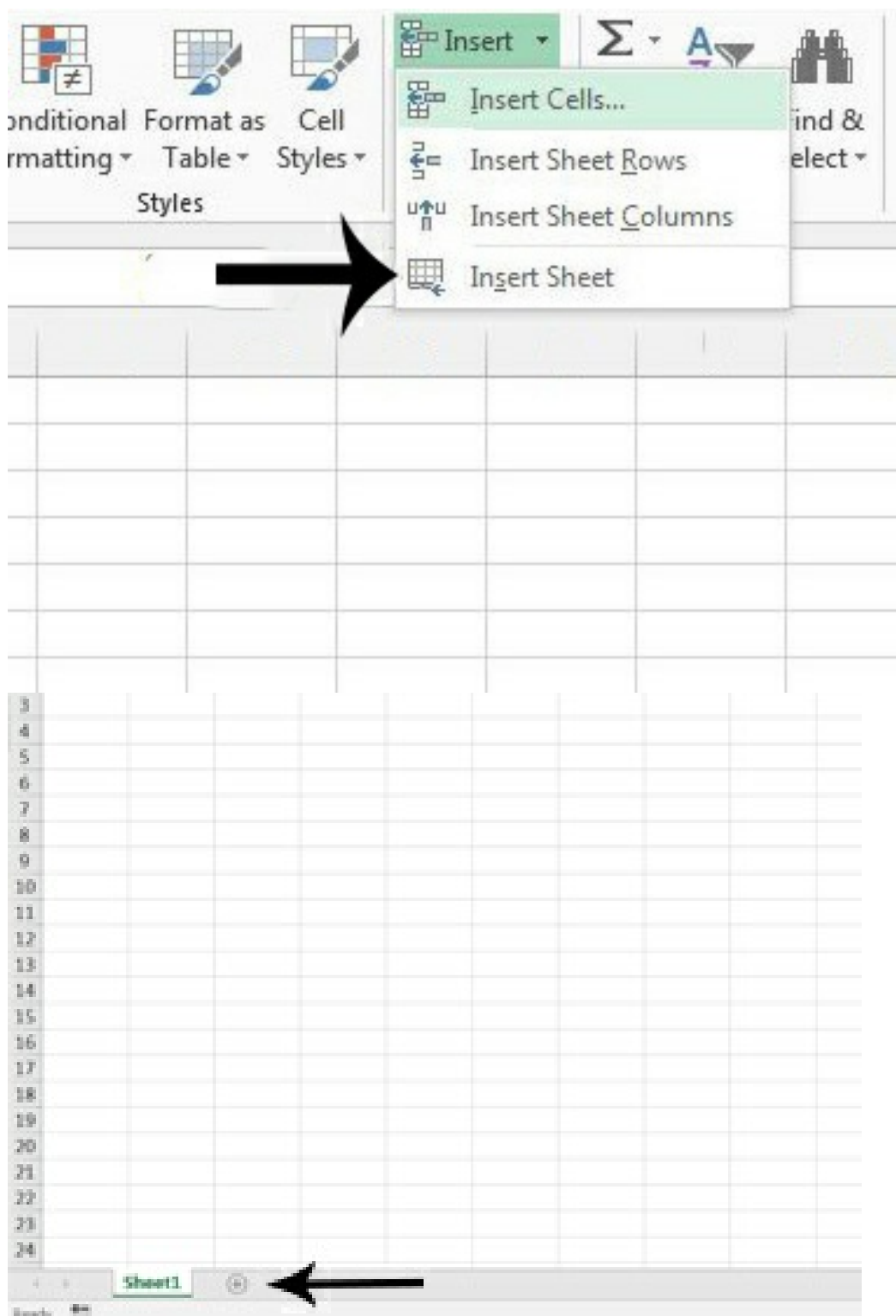
Spanishi modes: Tuteo verb forms only

Dictionary language: English (United States)

CHAPTER TWO- USING YOUR WORKSHEET

How to Create a New Workbook

- Doing this is very easy. First, open Excel, press Ctrl+N and then start inputting your data.
- To create a workbook from a template, click on File, click on New, select a template, and start inputting your data.
- To insert a worksheet, click on the Plus icon  next to the name of your worksheet at the bottom. Or, click on Home, go to Insert, and then click on Insert Sheet.



- If you prefer giving your worksheet a unique name, then double-click on the sheet's name to rename it.
- To delete a worksheet, right-click on the Sheet tab and then select Delete from the options. That's it!

Enter data

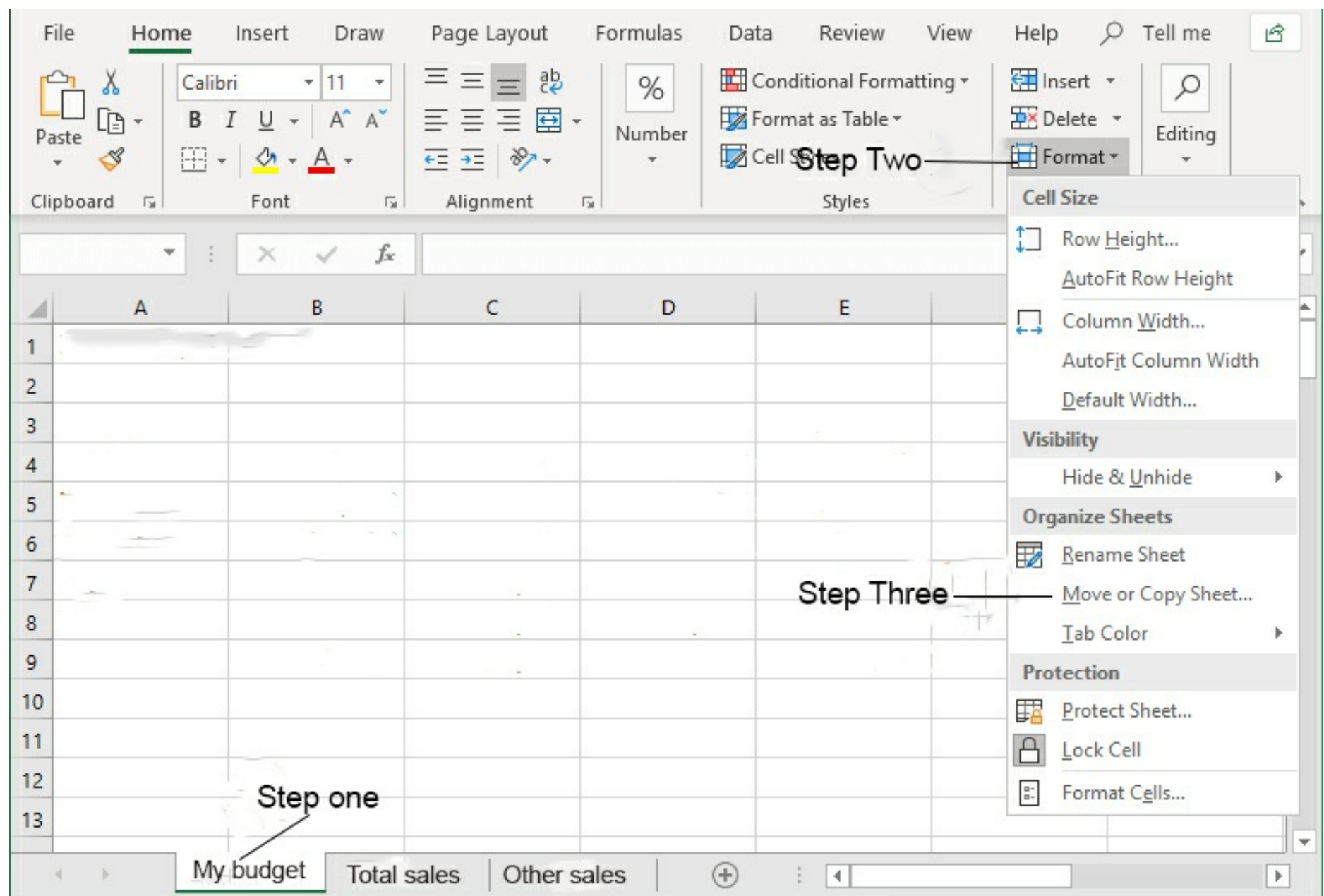
- Click on a cell, type in your data, and then press the ENTER key. Press ALT+ENTER on your keyboard to enter data on a new line in a cell
- To enter a date range, such as days, or months, enter the first value in a cell and enter the next value in the next cell to form a pattern.
- If you'd like to form a series with the data entered in the cell, insert the first two values in the first two cells and drag the cell. E.g., type A and B in the first two cells if you want the series A, B, C, D, E...,.
- Drag the fill handle of the cell containing the starting value. Fill the range by moving the handle across it. Drag down or to the right to fill in ascending order. Drag up or to the left to fill in descending order.
- If the series doesn't auto-fill, click on the box close to the cell's handle and then select "Fill Series."

The diagram illustrates a 10x10 grid with a 4x4 shaded area in the bottom-left corner. A green border highlights the shaded area and the first column of the grid. A black arrow points to a small icon in the bottom-right corner of the grid, which depicts a building with a lightning bolt.

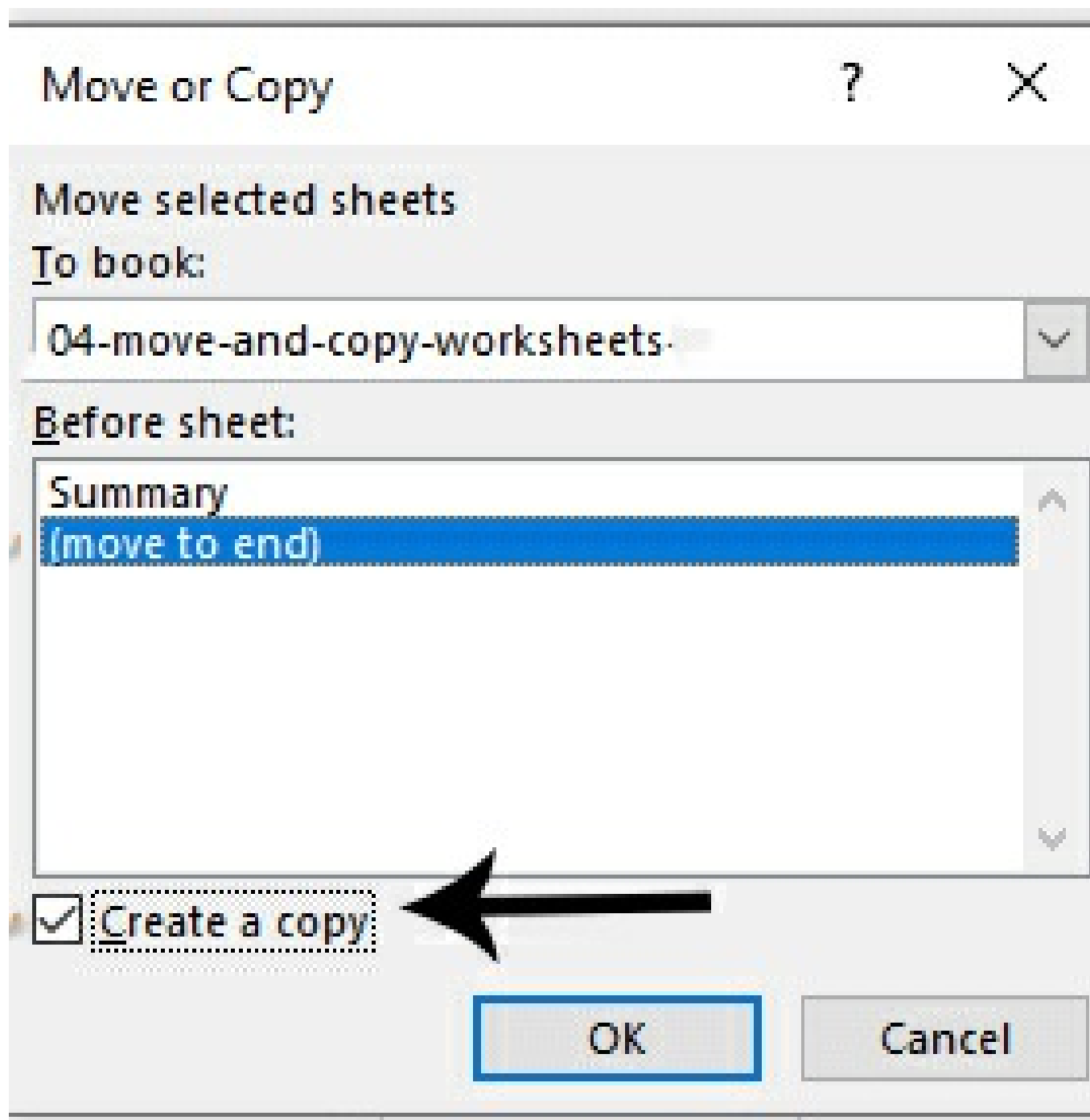
Copying or Moving a Worksheet

Excel allows you to copy or move your current spreadsheet even after creating it. You can use the “Move or Copy” dialog box to improve accuracy. With the steps below, you can export/save one or more worksheets.

- Head to the Home tab, click on “Format,” and then click on “Move or Copy Sheet.”



- Next, select the worksheet's location. Note that the workbook you wish to move the worksheet to must be open for it to appear in this list.
- Next, select a specific location where you want the sheet to appear in the selected workbook.
- Click on the "Create a copy" box and then click OK. If this check box is not selected, the worksheet will be deleted from its current location as it's been moved to the selected workbook. That's it!



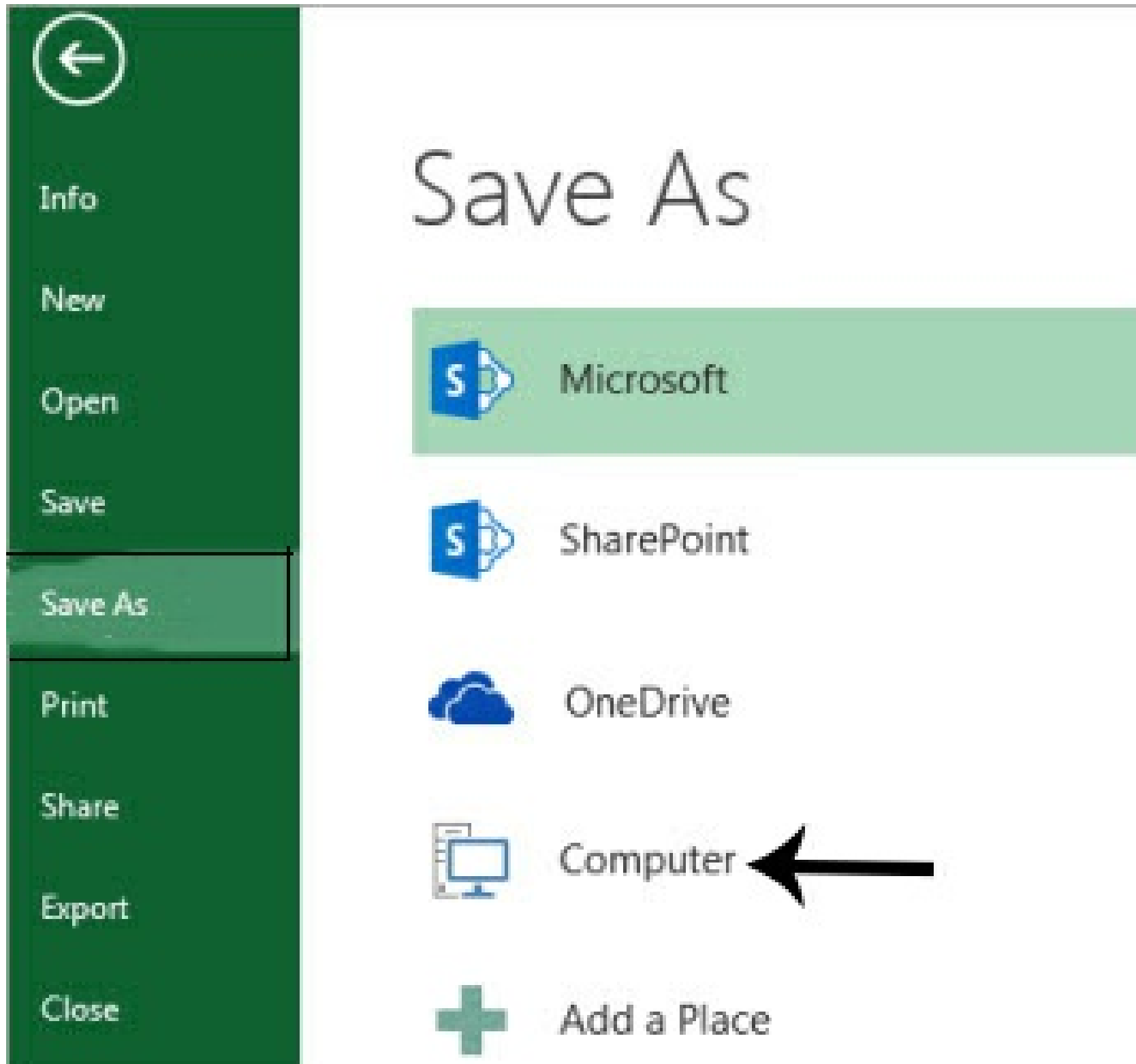
How to drag and drop the worksheet

- First, click on the sheet tab to select it, and then drag it to its new location. Release the mouse to set the sheet in its new location.
- Note that using this process to move your sheet won't leave duplicates behind when you are done.

Saving Your Document

You can easily save an existing document by using the shortcut "Ctrl+S," or you can follow the steps below:

- Click on File, click on Save As, and then select your preferred location to save the File. Or, you can click on "Add a Place" if you can't find your preferred location.
- If you selected "Computer" as your location, all you have to do is browse a preferred location to save the File.



How pin your favorite save location

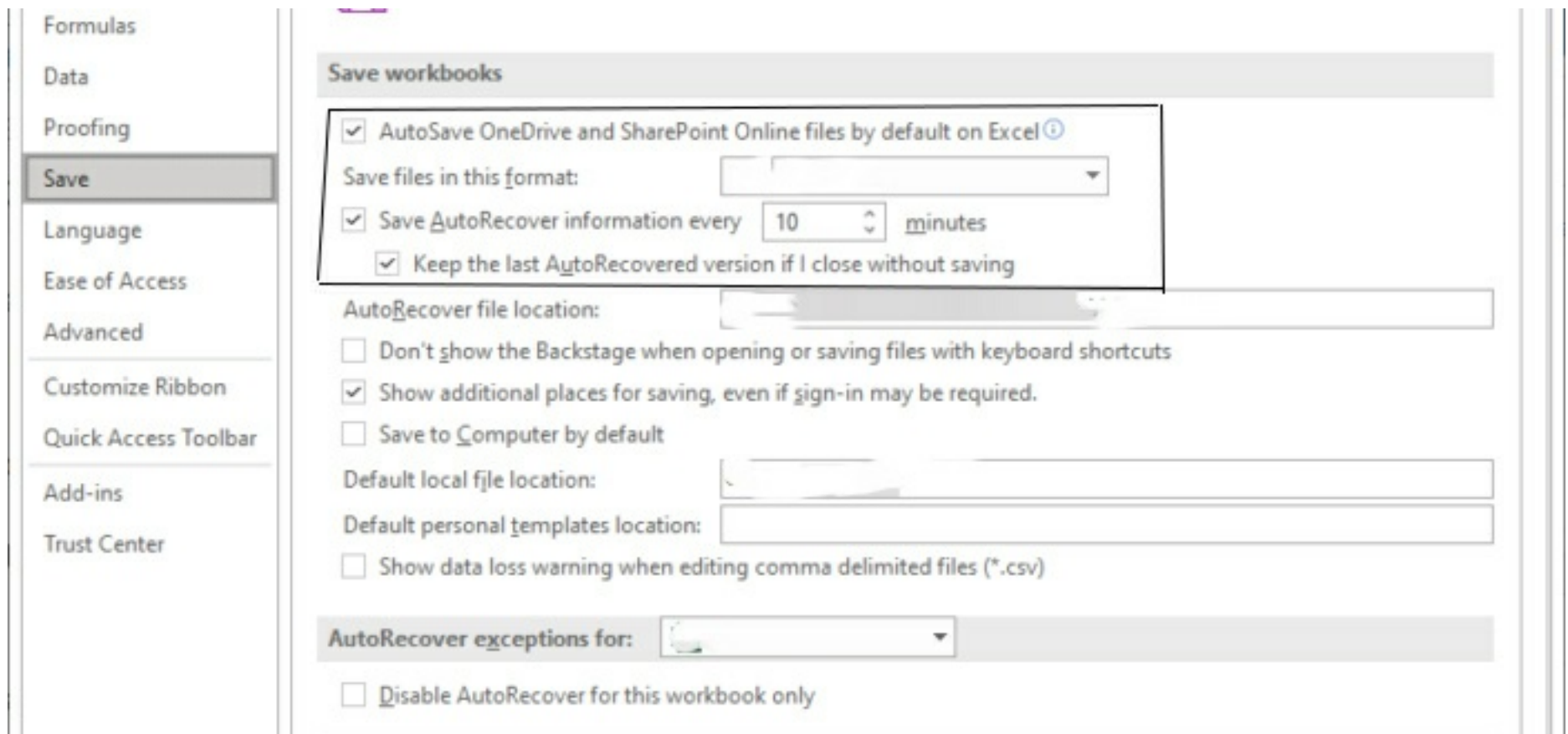
There is an option in Excel that allows you to "pin" or bookmark the position of your saved workbook. As a result, you'll have easy access to your workbook, and you'll be able to pin more than one workbook.

- Click on File, click on "Save As," and then select the location where you last saved your workbook.
- Look under "Recent folders" and then click on your preferred location.
- Next, click on the pushpin image that appears to pin that folder. The image will now display the Pin icon  .
- Click on the pin icon  again to unpin a location.

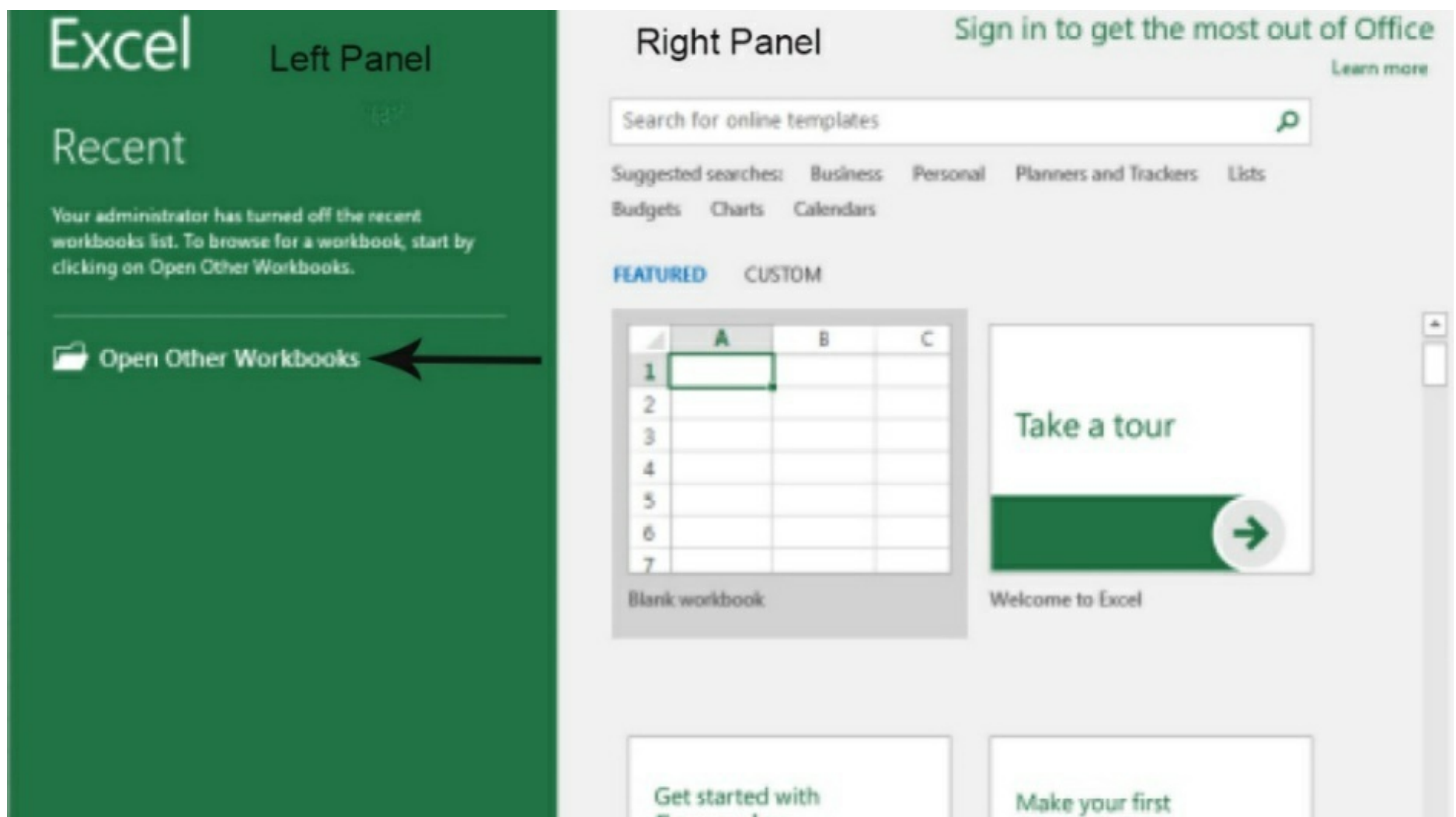
How to Recover Your Document Using Auto-save

Let's assume you are in a rush to go on a date and you weren't patient enough to save your work before clicking on the Exit button. Then you realize it will take hours to recreate your work. Fortunately, Excel can restore your files if the workbook was open for at least 10mins and if it created an Auto-Recover version. Here's how:

- First, make sure the autosave option has been enabled in settings.



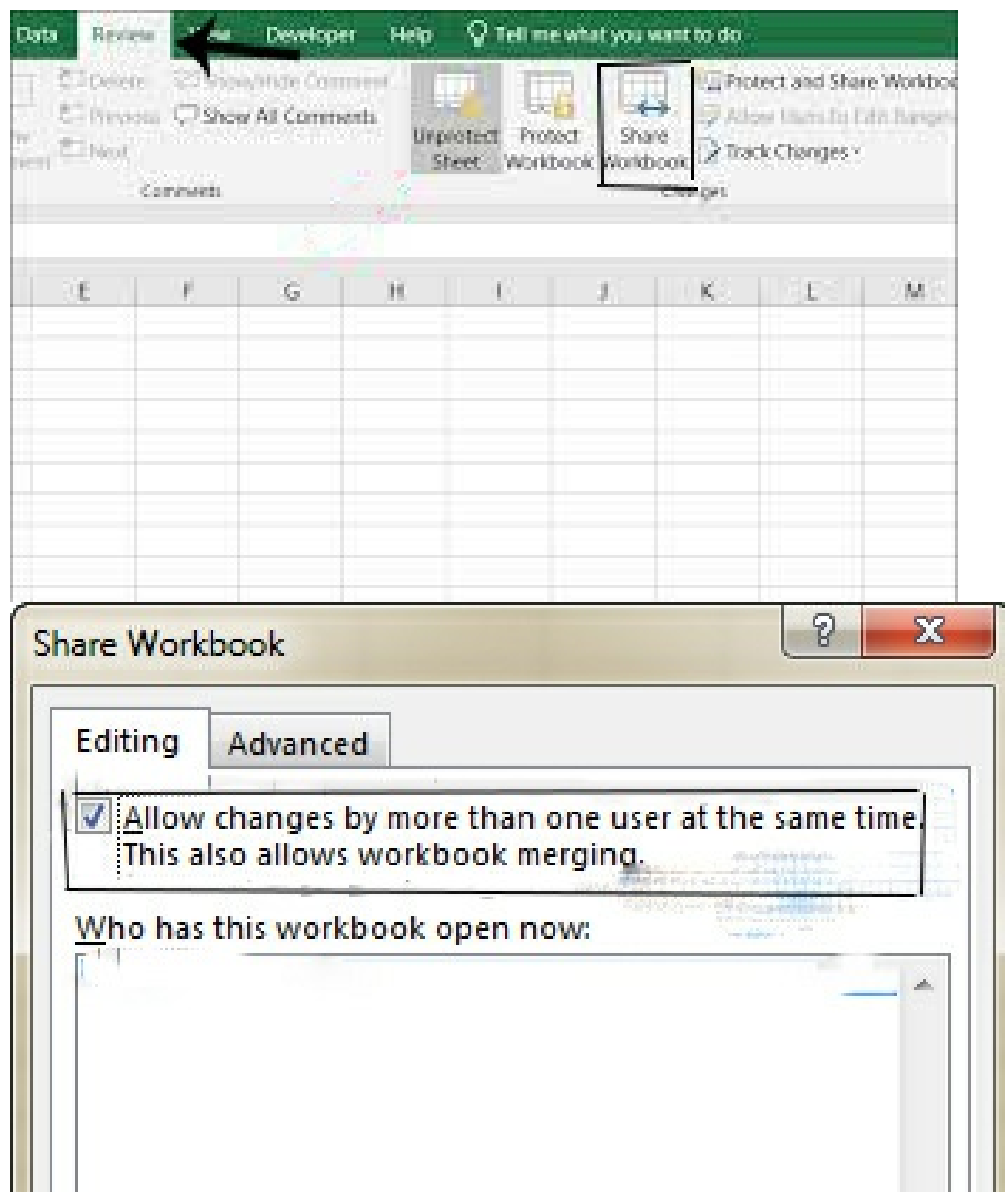
- Head to the Start screen, click on “Open Other Workbooks,” and then click on “Recover Unsaved Workbooks.” Here, you'll find all your unsaved workbooks that Excel recently saved



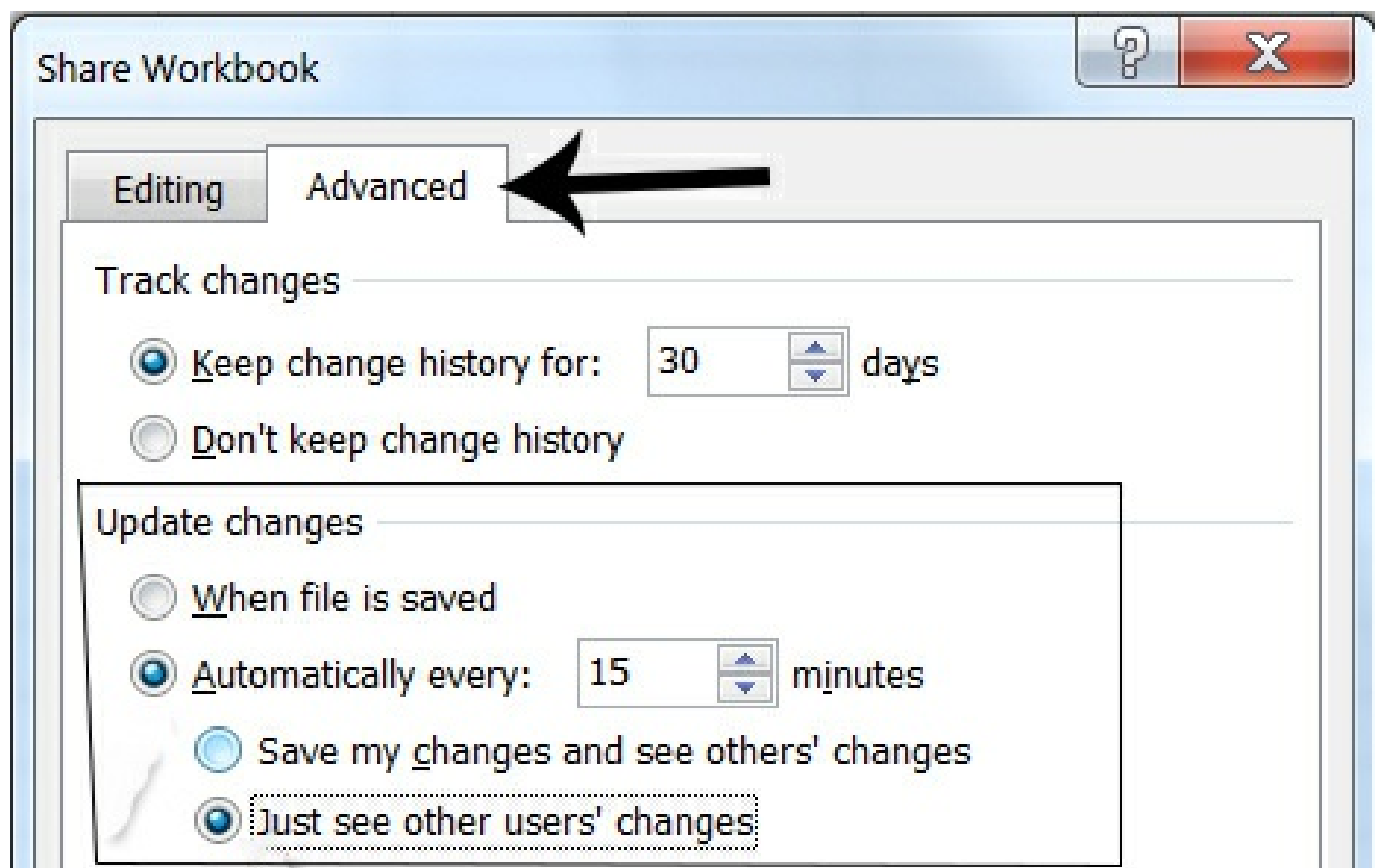
- Click on a workbook and then click on Open. Excel will automatically delete any unsaved workbook after 4 days.

How to share an excel workbook

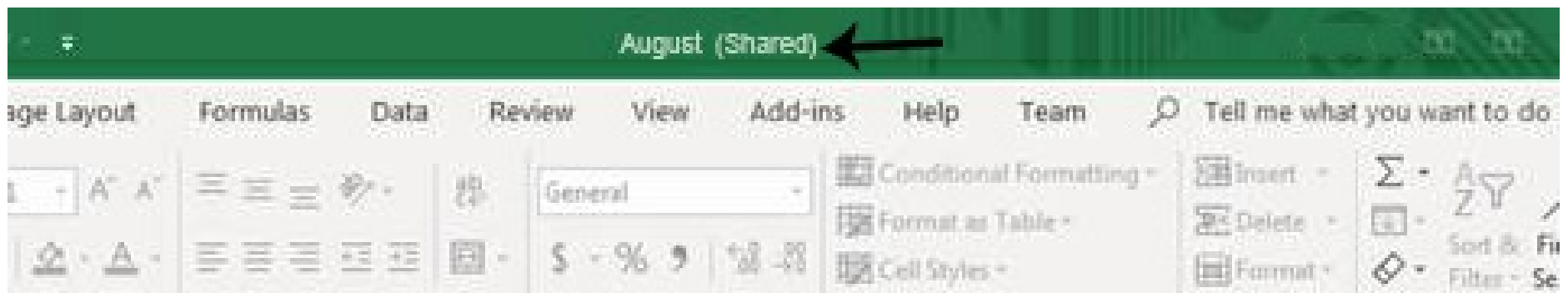
- Head to the Review tab, click on “Share Workbook,” and then check the “Allow changes by more than one user at the same time” box that appears on the screen. This will also allow you to merge the workbook’s check box on the Editing tab.



- Or, you can click on Advanced, select your preferred “track changes” settings, and then click OK.



- Next, press the shortcut “Ctrl + S” on your keyboard to save the Excel file. If done correctly, the word [Shared] will appear next to the workbook's name.



Now, you can share this Excel file with your friends or colleagues.

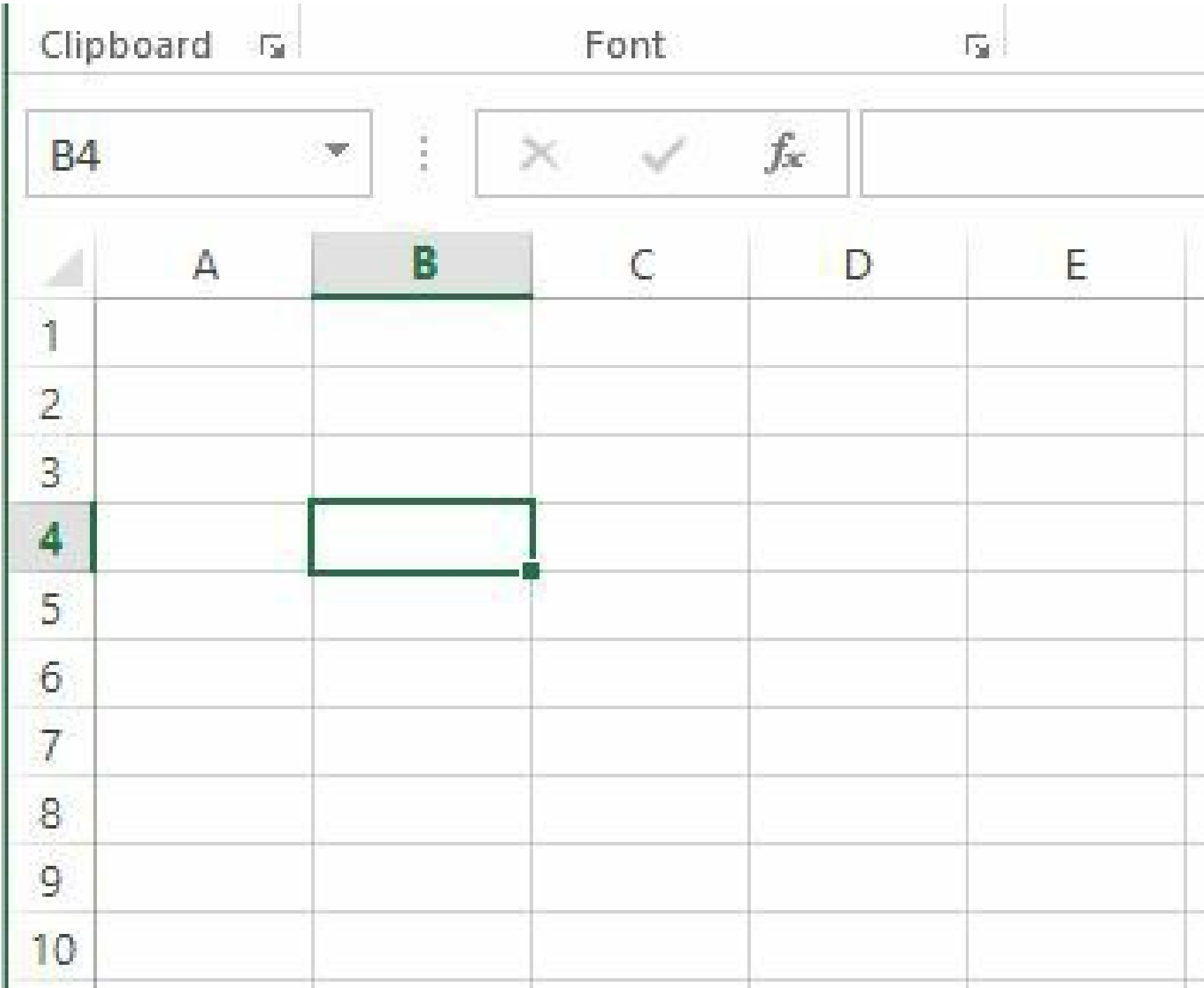
Note. If Microsoft Excel won't allow you to share a workbook, it's probably because:

- Tables and XML maps in workbooks can't be shared. Before sharing your Excel file, make sure to convert your tables to ranges and delete the XML maps.
- Some privacy settings are enabled. You need to turn them off.
 - Click on File, go to Options, go to "Trust Center," and then click on "Trust Center Settings"
 - Now, head to "Privacy Options" and then uncheck the "Remove personal information from file properties on save" option

CHAPTER THREE- CELL

Understanding the Cell

When working with Excel, you must enter data or content into cells. Columns are marked with letters (A, B, C, D), and rows are marked with numbers (1, 2, 3, 4). Each cell has its name or cell address, depending on the column and row. In the following example, the selected cell passes through column B and row 4, so the cell address is C4.



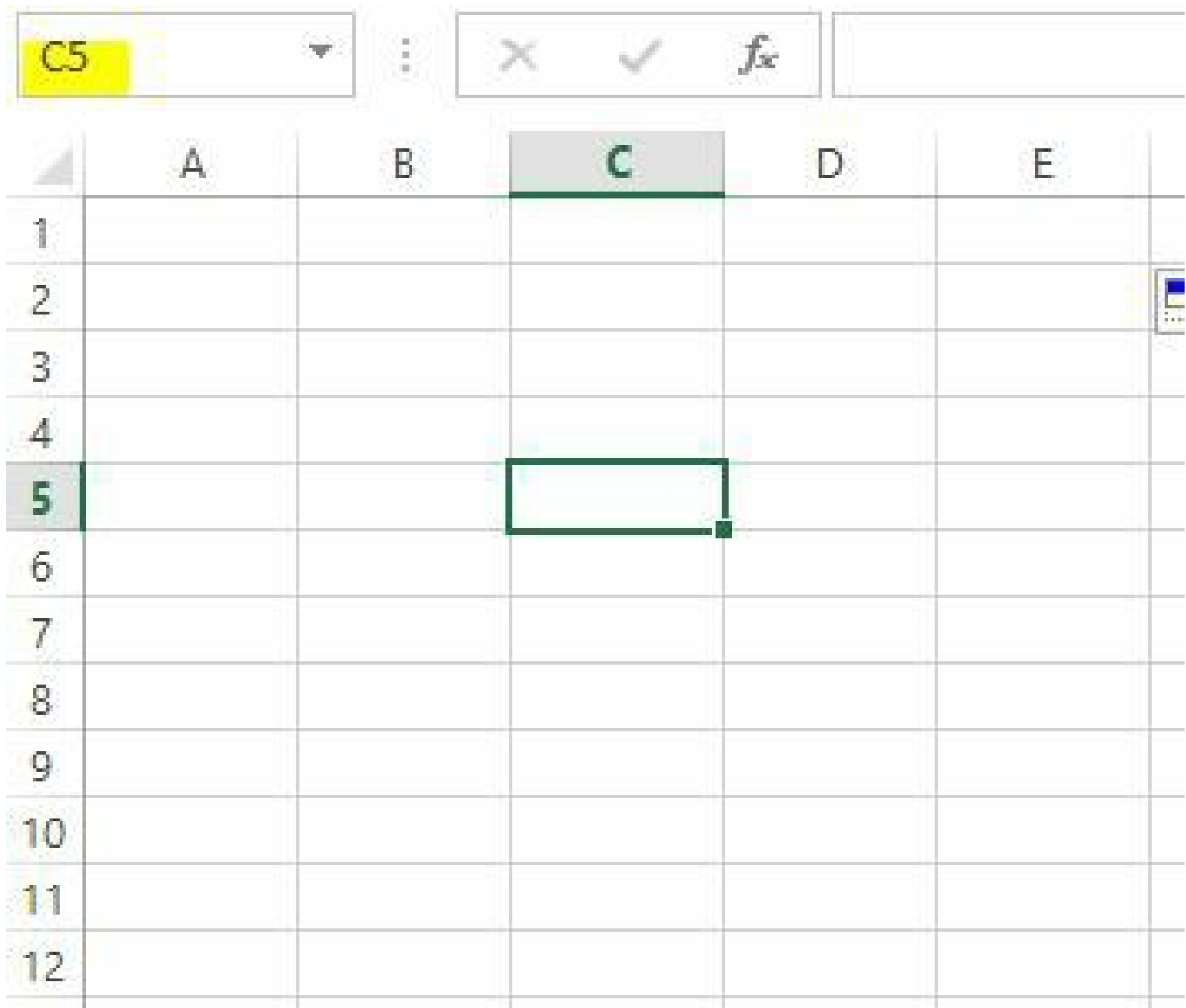
The cell address is also displayed in the name box at the top corner, and selecting a cell highlights the cell row and column headers.

You can also choose several cells at once. A cell range is a collection of cells. For example, If I select cells A2, A3, A4, down to A8, the cell range will be written as A2:A8. The screenshot below shows some examples

Cell Range A2:A8

How to select a cell and a cell range

- Click on a cell to select it and then input your data or edit the cell content. The selected cell will have borders around it. The column/row headings of the cell will be highlighted. I have selected the cell C5 in the example below.



Now, if you'd like to select a larger group of cells (cell range), here's how:

- Click on a cell, drag the mouse to highlight all cells you want to select, and then release the mouse. That's it! I used the cell range C1:G10 for this example.

Cell Content

Each cell contains text, formatting, formulas, and functions.

- Text: you can input letters, numbers, and dates into a cell

C4		:	✕ ✓ f_x		14
	A	B	C	D	E
1	Date	Budget	Sales		
2	4/8/2021	\$40	56		
3	5/8/2021	\$50	33		
4	6/8/2021	\$60	14		
5					
6					
7					
8					
9					
10					
11					
12					

- **Formatting attributes:** You can use formatting attributes in cells to change the appearance of characters, numbers, and dates. For example, a percentage can be written as 0.30 or just 30%. You can even edit the text color or the cell background color.
- **Formulas and functions:** these features are used to calculate cell values. In the example below, SUM(B2:B4) adds the values in each cell in the range B2:B4 and then displays the total in cell B5.

B5

⌵

:

✕

✓

f_x

=SUM(B2:B4)

	A	B	C	D	E	F
1	Date	Budget	Sales	Click here for Formula		
2	4/8/2021	\$40	56			
3	5/8/2021	\$50	33			
4	6/8/2021	\$60	14			
5	Total	\$150				
6						
7						
8						

How to insert content

- Select a cell, type your data and then press the Enter key on your keyboard. The content of the cell and the formula bar will be displayed. You can also use the formula bar to enter and edit cell content. I used cell C6 for this example, and I typed “Sports” into the cell.

FILE HOME FORMULAS PAGE LAYOUT

C6		✕ ✓ <i>fx</i>		Sport			
	A	B	C	D	E	F	G
1	Date	Budget	Sales	The formula bar			
2	4/8/2021	\$40	56				
3	5/8/2021	\$50	33				
4	6/8/2021	\$60	14				
5	Total	\$150					
6			Sport				
7							
8							
9							
10							

How to delete/clear cell content

- First, select the cell, then head to the Home tab and click on the Clear command.
- Next, click on “Clear Contents” to delete/ clear the cell contents. That’s it!

I used the cell range A3:C3 in the screenshot below

Clipboard

Font

A3

:

X

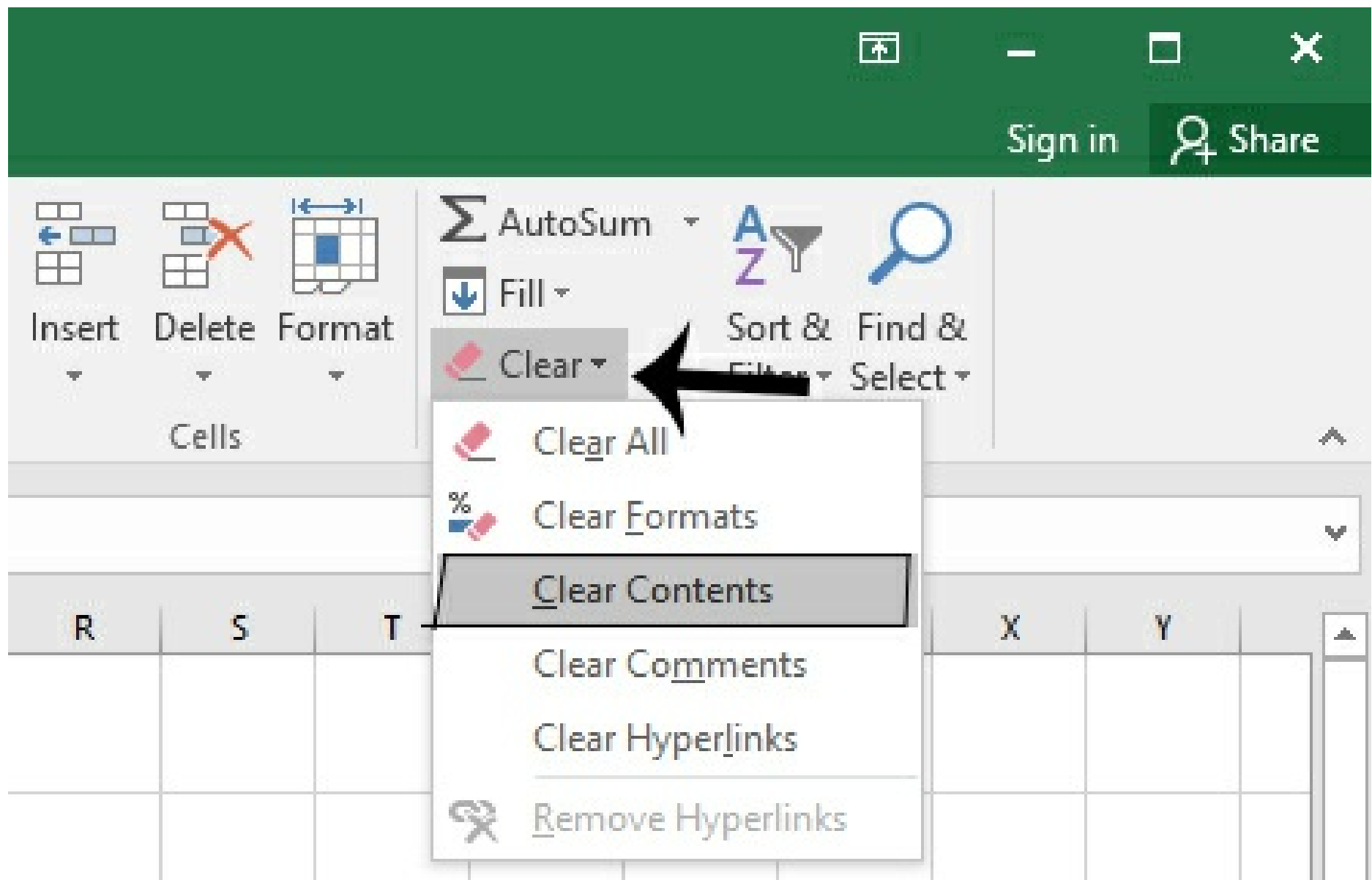
✓

fx

5/8/2021

	A	B	C	D	E
1	Date	Budget	Sales		
2	4/8/2021	\$40	56		
3	5/8/2021	\$50	33		
4	6/8/2021	\$60	14		
5	Total	\$150			
6					
7					
8					
9					
10					
11					

Select the cell

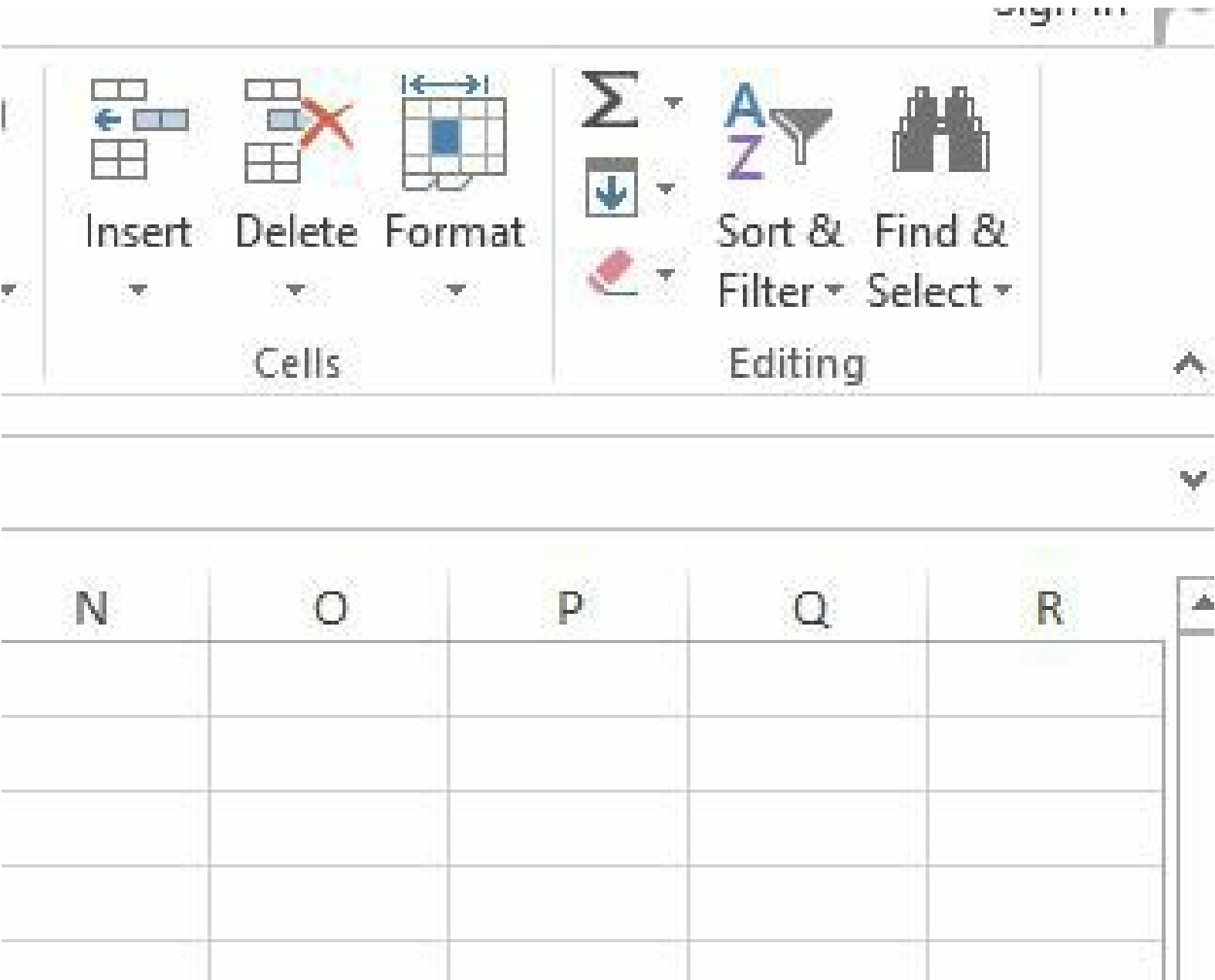


- Press the “Delete” key to delete multiple cells at once. Press the Backspace key to clear the contents of one cell at once

How to delete cells

When you delete an entire cell, the cells below it will move to replace the deleted cell.

- Select the cell(s), head to the Home tab, and then click on the Delete command. The cells below will replace the deleted cell.



How to copy and paste cell content

- Excel allows you to copy and paste data from one cell to another, saving you time and effort.
- Select the cell(s), head to the Home tab, and click on “Copy.”
- Or, use the keyboard shortcut Ctrl+C. I used cell C6 for this example.

The screenshot shows the Microsoft Excel interface. The ribbon is set to the 'Home' tab. The 'Clipboard' group includes 'Cut', 'Copy', and 'Paste' buttons. The 'Font' group includes font face (Calibri), font size (13), bold (B), italic (I), underline (U), and text color options. The spreadsheet below has columns A (Date), B (Budget), C (Sales), D, and E. Row 6 is highlighted, and cell C6 contains the text 'Sport'. A dashed green box surrounds cell C6, indicating it is the source of the copy operation.

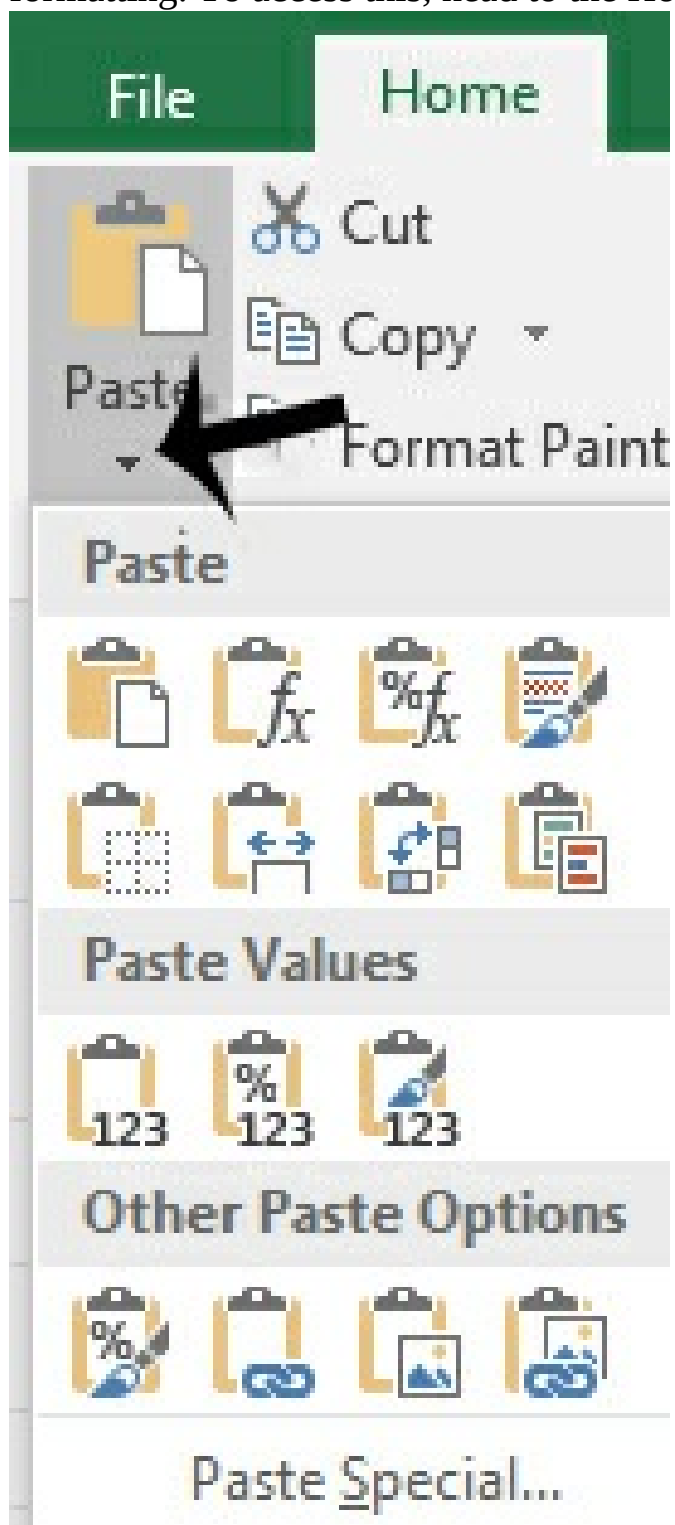
	A	B	C	D	E
1	Date	Budget	Sales		
2	4/8/2021	\$40	56		
3	5/8/2021	\$50	33		
4	6/8/2021	\$60	14		
5	Total	\$150			
6			Sport		
7					
8					
9					
10					
11					
12					

- Next, select the cell(s) where you want to paste the content and then press the shortcut Ctrl+V on your keyboard or click on the Paste command in the Home tab. That's it
- I used cell C9:C14 for this example. A dashed box will surround the duplicated cell(s).

5	Total	\$150		
6			Sport	
7				
8				
9			Sport	
10			Sport	
11			Sport	
12			Sport	
13			Sport	
14			Sport	
15				 (Ctrl) ▾
16				

How to access more paste options

You can also use other paste options, which can be pretty helpful when working with cells that contain formulas or formatting. To access this, head to the Home tab and click on the drop-down arrow next to the Paste command



Instead of clicking on the ribbon to select a command, you can quickly access commands by right-clicking them. Just right-click the mouse and then select the cell you want to format. You will see a drop-down menu with various commands that are also available on the ribbon.

How to cut and paste cell content

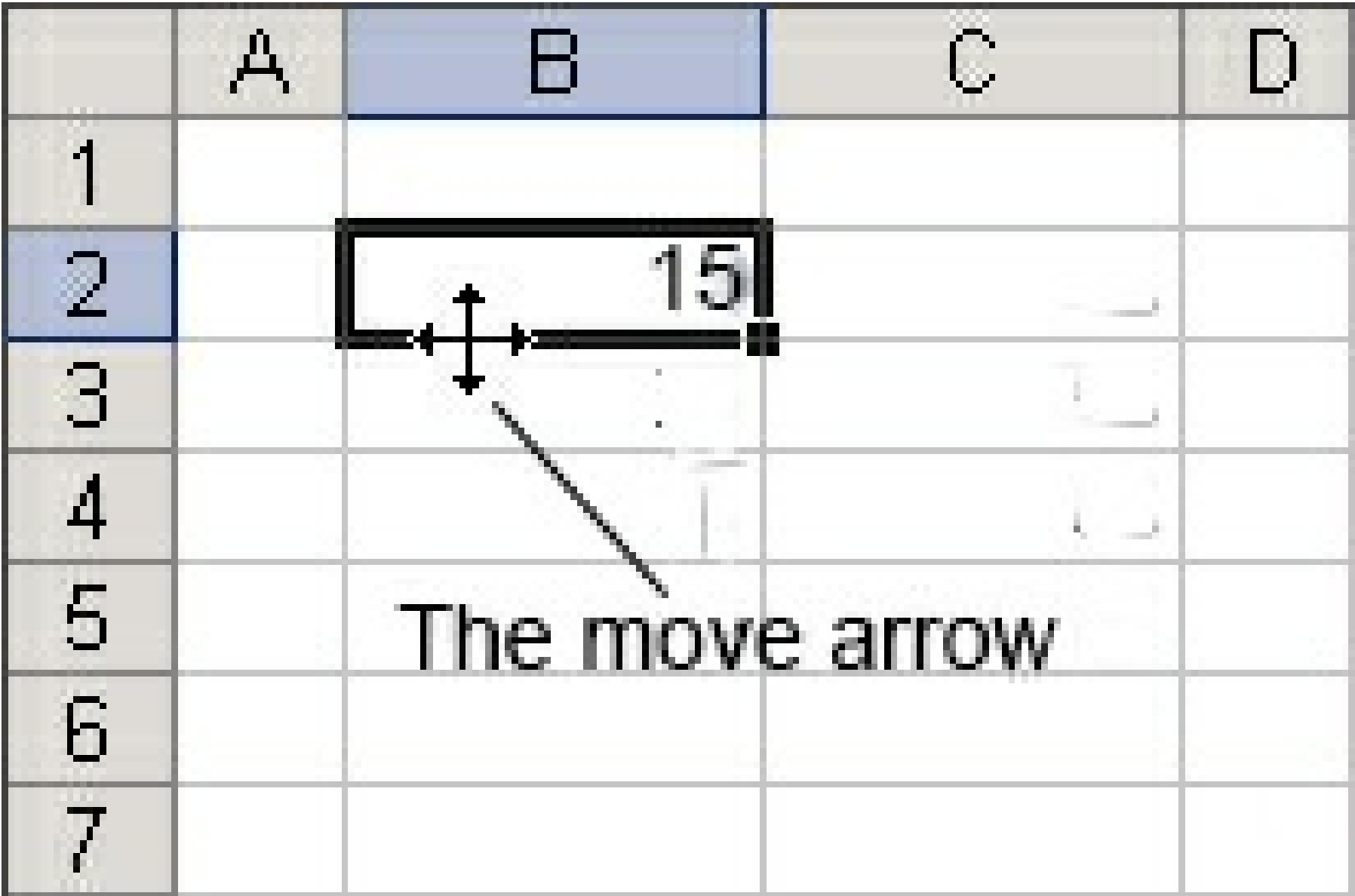
This feature allows you to move content between cells. Here is how:

- Select the cell(s), press “Ctrl+X,” or right-click the mouse and select the Cut command.
- Next, click on the cells you want to copy the content into and then use the shortcut Ctrl+V or right-click on the mouse and select the Paste command. That’s it.

How to drag and drop cells

The drag and drop feature allows you to move contents within a cell without cutting or copying them.

- Select the cell(s), and then move the mouse to the edge of the cell box until the mouse changes to 4 pointed arrows.

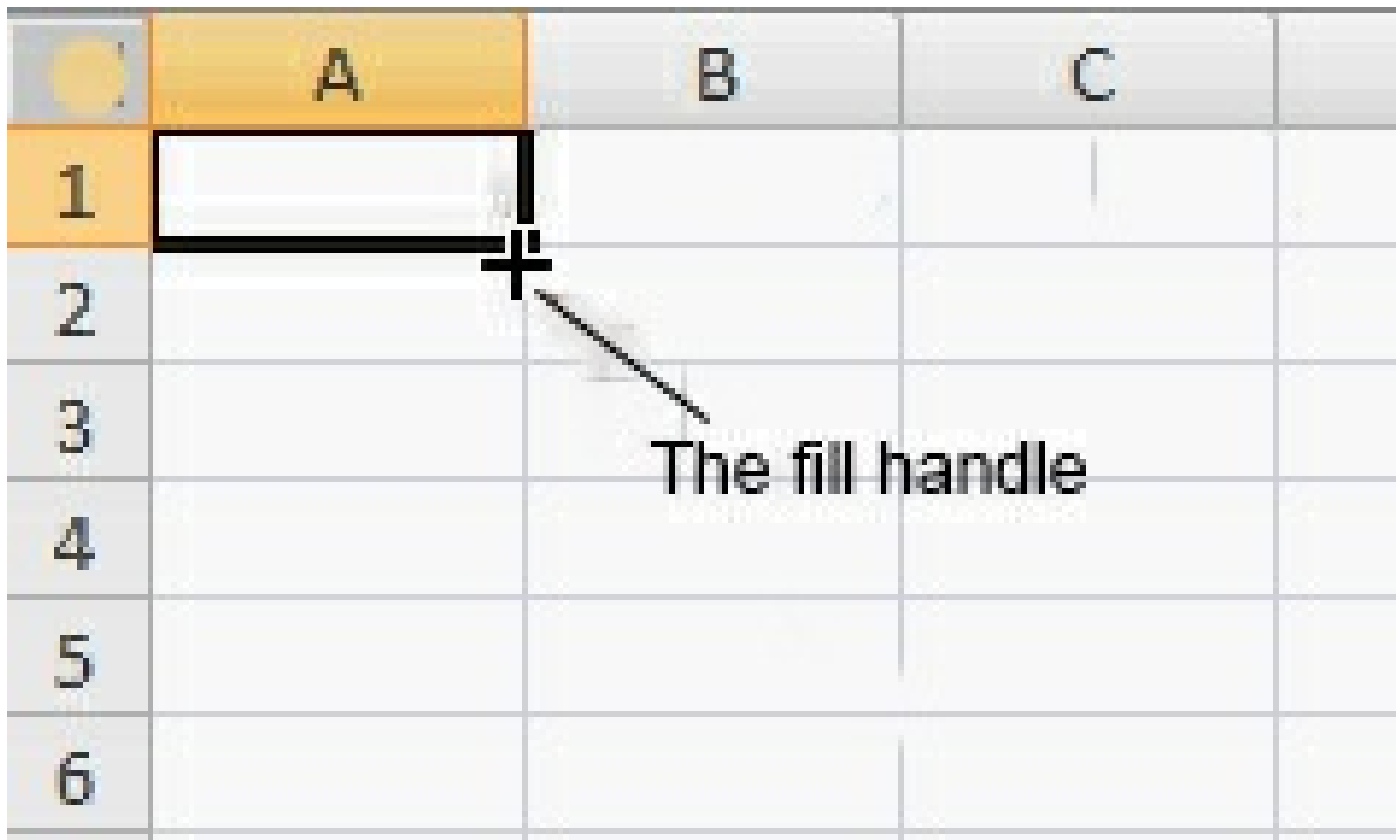


- Click and drag the cells to a new location.
- Release the mouse to drop the content of the cell in its new location.

How to use the fill handle

Let's say you typed 15 into a cell, and then you want to auto-fill the number 15 in another adjacent cell. Using the fill handle is another good alternative

- First, select the cell(s) and then move the mouse to the edge of the cell box until the fill handle appears.



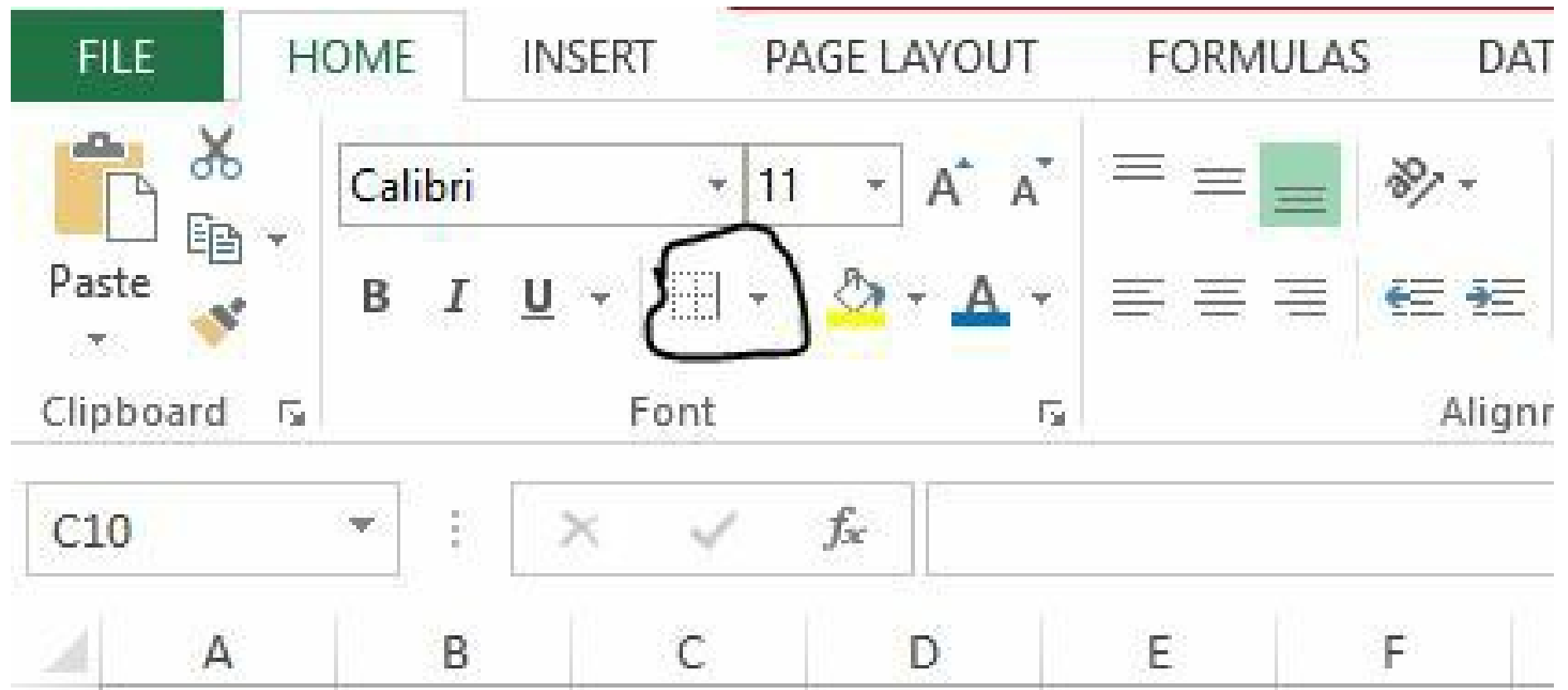
- Select the cells you want to fill by clicking and dragging the fill handle and then release the mouse to fill the selected cells.

Cell Borders

You can add and remove borders from your cell. To give your borders a new look, you can add styles, thickness, colors and change the layout of your borders.

How to add a cell border

- First, select the cell(s), and then head to the Font group in the Home tab.
- Tap on the drop-down arrow next to Border  and then select your preferred border style.



- If you don't find your preferred border style, then click on "More Borders," look under "Line and Color," and then click on your preferred line style and color.

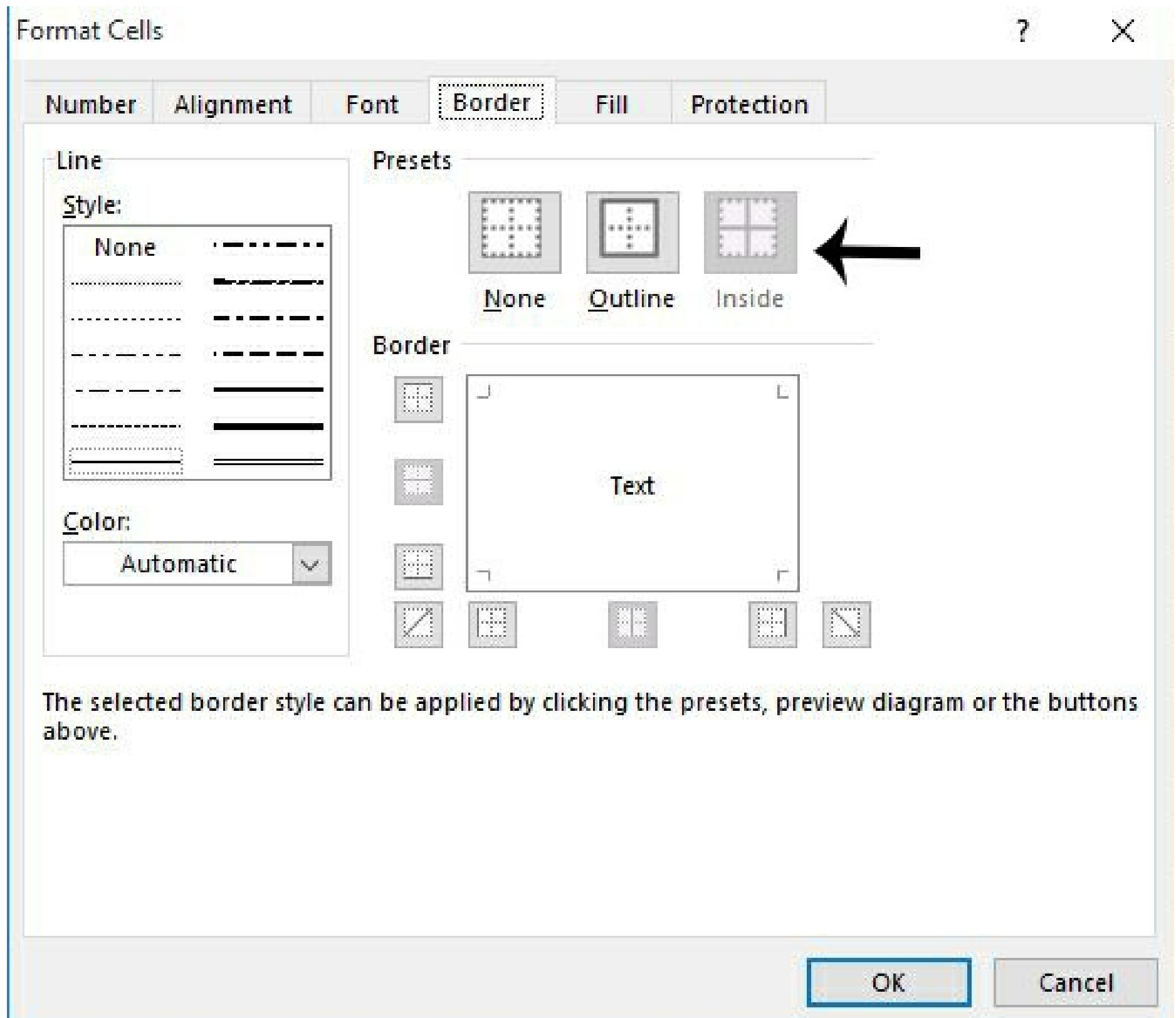


- You can also click on “Draw Border” and then use the pen to draw borders around the cell



If a page break separates a cell and you need to apply an inside border on a border that appears on only one page. Here's how to do it:

- Select the rows on both sides of the page break, click on the drop-down arrow next to Border and then click on “More Borders.”
- Next, click on the Inside button  under “Presets,” look under Border , and then click on the vertical Border to remove it.



Modifying the Column, Row and Cell In the Worksheet

The rows/columns of your new workbook have the same height and width. Fortunately, you can edit them, which also includes wrapping textual content and merging cells.

How to modify column width

- First, click on the column header (e.g., click on column A), and then hover the mouse over the column until the mouse pointer changes to a double-arrow. Note, a thin line will appear when you are increasing/decreasing your row/column.
- Next, click and then drag the mouse to increase or decrease the column width. The column's width will change when you release the mouse.
- If the column is too tiny for numerical values, the cell will display pound signs. Increase the column's width to make the data visible.

	A	B	C
1			
2			
3			
4			
5			
6			
7			

First click here

Drag this arrow to resize

	A	B	C	D
1				
2				
3				
4				

Your cell should look like this:

	A	B	C
1			
2			
3			
4			
5			
6			
7			

How to AutoFit column width

The AutoFit feature allows you to set a column's width to fit its content automatically.

- Move the mouse pointer to the column heading and then position the pointer to the column row until the cursor turns into a double arrow.
- Next, double-click the mouse to fit the column's content automatically.
- You can also automatically adjust the width/height of multiple columns/rows at the same time. Select the columns, head to the home tab, and click on "AutoFit Column Width."

How to modify row height

- Move the mouse pointer over the row line and position it until the cursor becomes a double arrow.
- Next, click and then drag the mouse to increase or decrease the row height. That's it!

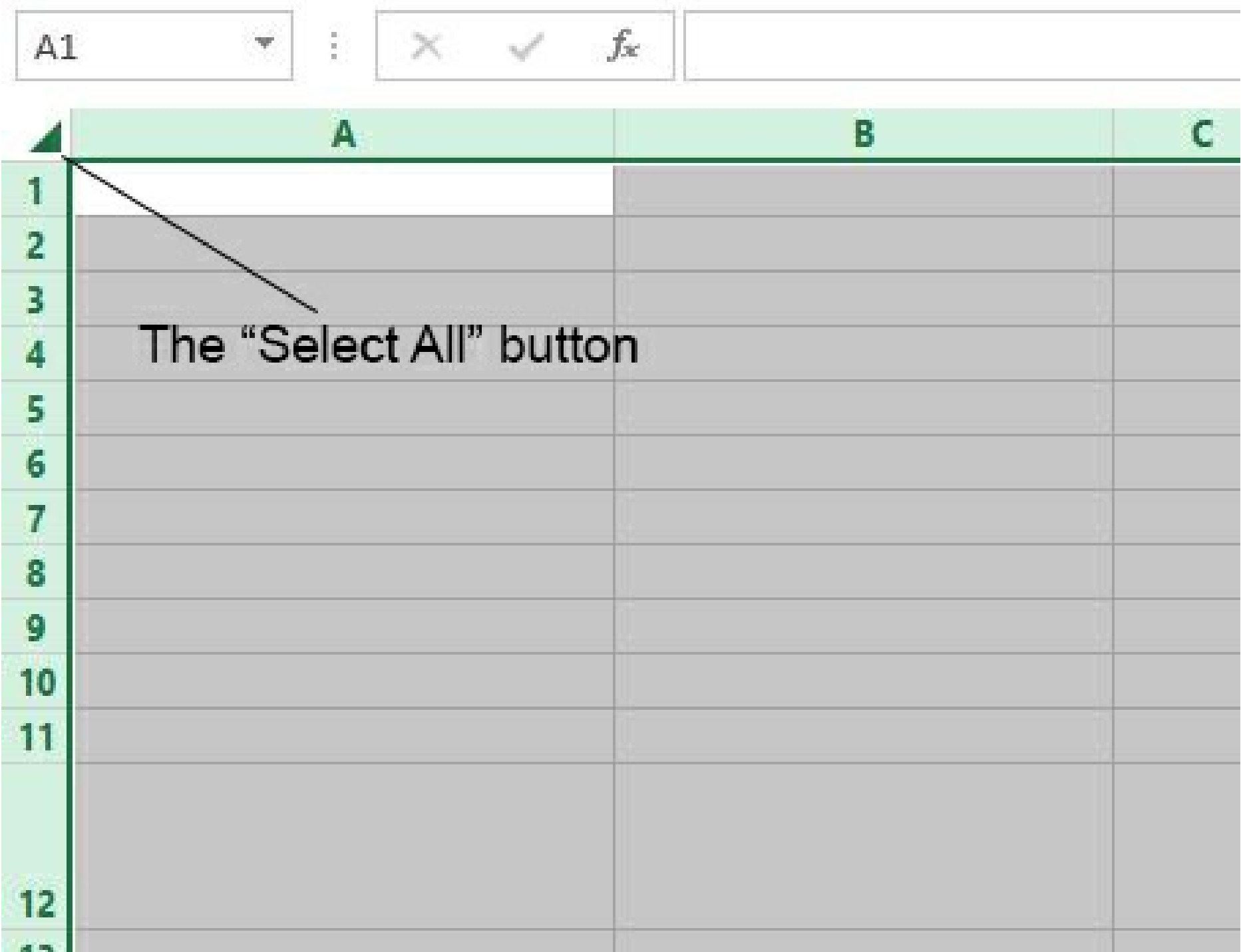
Your cell should look like this:

9					
10					
11					
12					
13					
14					
15					
16					
17					
18					

How to modify all rows or columns

You can change the height and width of each row and each column simultaneously instead of adjusting the size of the rows and columns individually. This way, you can easily set a uniform size for each row/column in your worksheet.

- First, click on the “Select All” button to select every cell in the worksheet.
- Move the mouse over the row line and then position it until the cursor becomes a double arrow.
- Next, click and drag the mouse to increase or decrease the row height for the entire worksheet.



Inserting and Hiding Cells

How to insert rows

- Let’s say I need a new row between rows 6 and 7. So, I’ll select row 7 for this example. Select the row heading below where you want the new row to appear, head to the Home tab, and click on Insert. That’s it
- A paintbrush icon will appear next to the newly inserted cells. Use this button to change the formats of these cells. To access other options, hover your mouse over the icon, then click the drop-down arrow.

3					
4					
5	John	Doe			
6	Grace	Bull			
7	Jessie	Pan			
8	Jeremy	Sting			
9					
10					
11					
12					
13					
14					
15					

NormalBadGoodNeutral

InsertDeleteFormat

Σ AutoSum

↓ Fill

✖ Clear

A Z

Sort & Filter

Find & Select

3					
4					
5	John	Doe			
6	Grace	Bull			
7					
8	Jessie	Pan			
9	Jeremy	Sting			
10					
11					
12					
13					

- Repeat the same steps to insert a new column.
- Note that you must select the entire row or column when inserting new rows and columns. Simply click on the row/column heading to select them. The Insert command will only insert a new cell only if a cell in the row or column is selected.

How to hide and unhide a row/column

Select the columns/rows you want to hide, right-click the mouse, and then click on Hide. That's it! The green line indicates the hidden column

[illegible]

	A	B	D	E	F	G
1						
2		Column C is gone				
3						
4						
5	John	Doe	dog			
6	Grace	Bull	cat			
7	Jessie	Pan	pig			
8	Jeremy	Sting	horse			
9						
10						
11						
12						

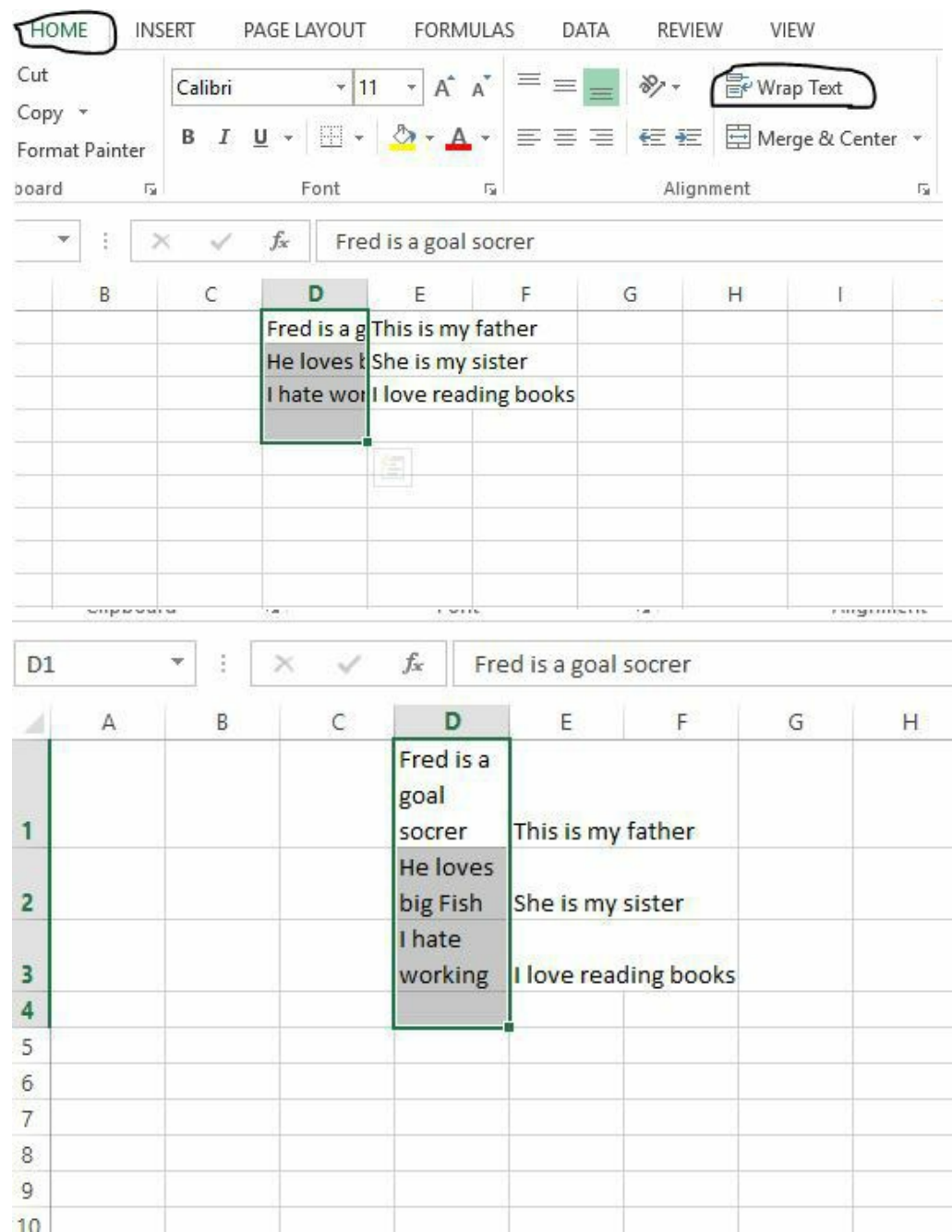
- Now, let's say I chose to hide column D, to unhide it, select columns C and F, right-click on the mouse, and then click on Unhide. The hidden column should appear on the list.

Wrapping Text and Merging Cells

If a single cell is filled with too much cell content, you can wrap the text or merge the cell rather than resizing the column. Wrapping text changes the row height, allowing cell contents to be displayed on several lines. Merging allows you to combine two or more empty cells into a single large cell.

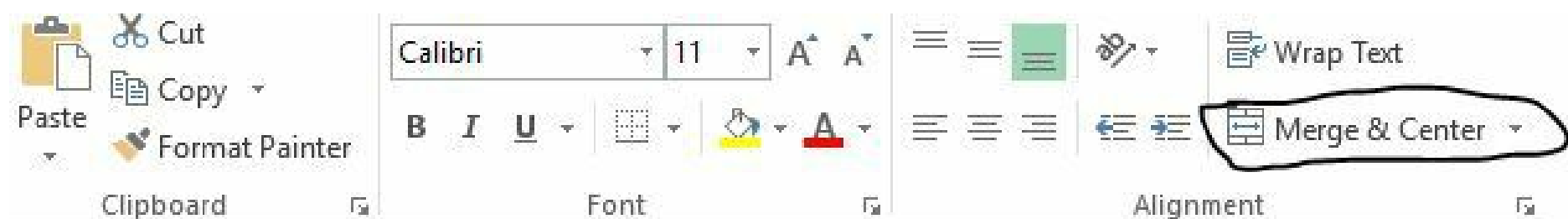
How to wrap text in cells

- First, select the cells, head to the Home tab, and then click on “Wrap Text.” This will wrap the text in the selected cells.
- Click on “Wrap Text” again to unwrap the text.



How to merge cells using the Merge & Center command

- Select the cell range you want to merge, head to the Home tab, and then click on “Merge & Center”
- The text will be centered and the selected cells will be merged. I used cells A1:G1 for this example



	A	B	C	D	E	F	G	H	I
1	Department of History logbook								
2				He loves	She is my sister				
3				I hate	I love reading books				
4				working					
5				Fred is a					
6				goal					
7				socrer	This is my father				
8									

	A	B	C	D	E	F	G	H	I	J
1	Department of History logbook									
2				He loves	She is my sister					
3				I hate	I love reading books					
4				working						
5				Fred is a						
6				goal						
7				socrer	This is my father					
8										

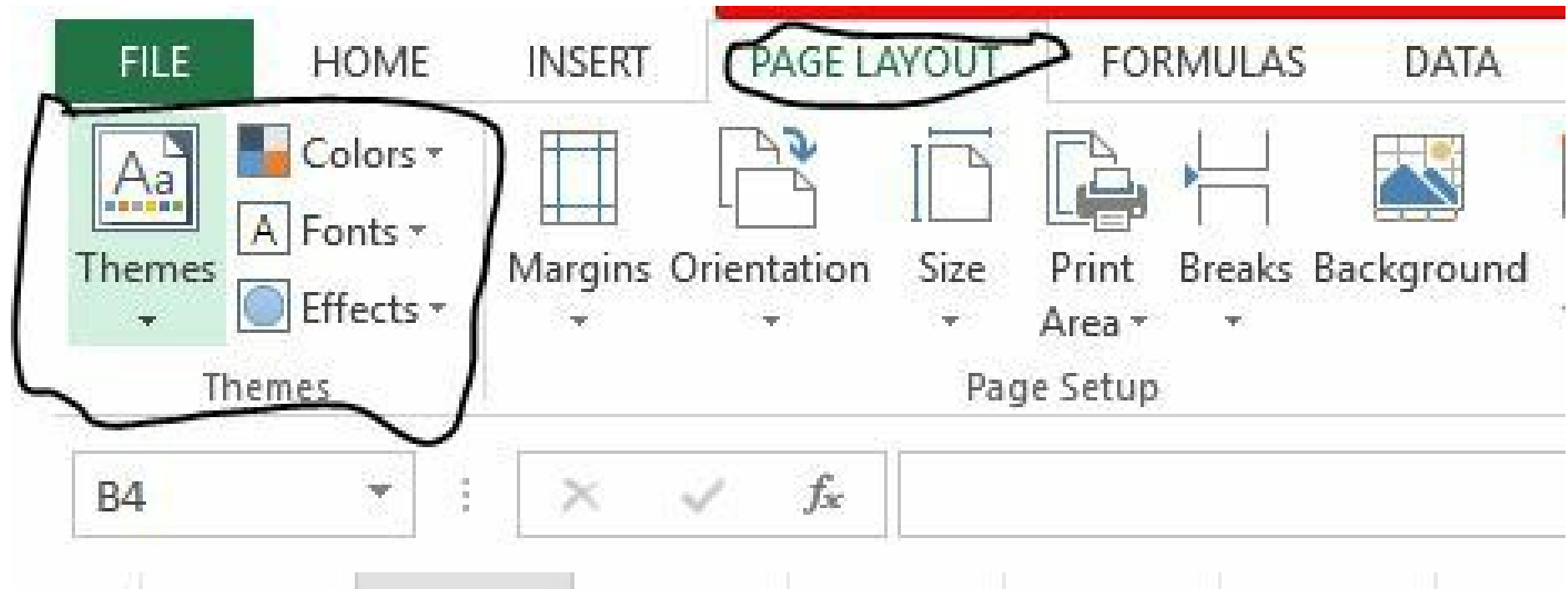
If you'd like to access other merge options, then click on the drop-down arrow next to the "Merge & Center." From here, you can choose from any of the following commands:

- Merge & Center: This will center your text and merge the selected cells into one cell.
- Merge Across: This will simultaneously merge the selected cells into larger cells and keep each row separate
- Merge Cells: This will only merge the selected cells into one cell.
- Unmerge Cells: This will unmerge the selected cells.

How to Change the Worksheet Theme Color and Effects

When changing colors, note that your color choice will affect the available colors in the color picker and the colors you've used in your workbook. If you'd like to give your worksheet a new look, here's how to do that

- First, head to the "Page Layout" tab, click on Themes and select your preferred theme. You can change your selected theme's fonts, colors, and effects as needed. You can also make it your default theme for all your new workbooks.



- Next, click on Colors to change the theme color and then select your preferred color set.
- Click on "Customize Colors" to create a new color set.
- Click on "More Colors" to create a new color and then enter color numbers on the Custom tab or select your preferred color on the Standard tab.
- Next, click on the name box, type a name for the new color set, and click on Save.
- You can check the Sample box for a preview of all your changes.
- Now to save all your color changes, head to the "Page Layout" tab, click on Themes, and then click on "Save Current Theme."

How to change theme fonts

Changing the theme font allows you to change all of your text at once. To make this work, make sure your text is formatted with Body and Heading fonts.

- Head to the "Page Layout" tab, click on Fonts, select your preferred font set, and click on "Customize Fonts" to create a new font set. Make sure you save your new font when you are done.
- Now to save your customized theme fonts, head to the "Page Layout" tab, click on Themes and then click on "Save Current Theme."

How to change theme effects

- Head to the "Page Layout" tab, click on Effects and select your preferred effects. Note that you can't customize a set of effects.

CHAPTER FOUR- FORMULAS AND FUNCTIONS

Basic Terms in Excel

For beginners to master financial analysis, they must first understand the basic Excel formulas. Microsoft Excel is widely regarded as the industry standard for data analysis software. Microsoft is one of the most popular companies among investment bankers and financial analysts in terms of data processing, financial modeling, and spreadsheets.

There are two main calculation methods in Excel: formulas and functions.

A formula is an expression that works with values in a cell(s). For example, =C1+C2 +C3 calculates the sum of the values in cells C1 through C3.

Functions are predefined formulas. They do away with the time-consuming manual entering of formulas by providing human-friendly names. For instance, the Function =SUM will command excel to add up all of the cells' values from C1 to C3.

Formulas practical exercise

Now let's look at an easy example using Formulas:

- Open the Excel program, create a new workbook, and then enter the data shown in the image below

Clipboard		Font		Alignment			
E4	:	X	✓	<i>f_x</i>			
	A	B	C	D	E	F	G
1	Car shop budget						
2							
3	S/N	Item	QTY	Price	Total		
4	1	Tires	10	\$400			
5	2	Engines	11	\$500			
6	3	Brake Pads	13	\$600			
7	4	Air compressor	10	\$700			
8	5	Oil Drain	2	\$800			
9							
10							
11							
12							
13							
14							

- Now, let's write the formula that calculates the subtotal. Set the focus to cell E5 and then enter the following formula "=C5*D5"
- Here, "C5*D5" multiplies the values of cell addresses C5 and D5 using the arithmetic operator multiplication (*).

Clipboard		Font		Alignment			
SUM		✕ ✓ <i>f_x</i>		=C5*D5			
	A	B	C	D	E	F	G
1	Car shop budget						
2							
3	S/N	Item	QTY	Price	Total		
4		1 Tires	10	\$400			
5		2 Engines	11	\$500	=C5*D5		
6		3 Brake Pads	13	\$600			
7		4 Air compressor	10	\$700			
8		5 Oil Drain	2	\$800			
9							
10							
11							
12							
13							
14							
15							

- Press the Enter key. You will get the following result

	A	B	C	D	E	F
1	Car shop budget					
2						
3	S/N	Item	QTY	Price	Total	
4		1 Tires	10	\$400		
5		2 Engines	11	\$500	\$5,500	
6		3 Brake Pads	13	\$600		
7		4 Air compressor	10	\$700		
8		5 Oil Drain	2	\$800		
9						
10						
11						
12						

- Now, select all the cells in column E to get the total using the same formula

Time-Saving Ways to Insert Data into Excel

There are five typical ways to insert basic Excel formulas while evaluating data. Each method has its advantage.

1. Simple insertion: Typing a formula inside the cell

Pasting basic Excel formulas is as easy as entering a formula in a cell or formula bar. Usually, it starts with an equal sign, followed by the name of the Function.

Excel will display a function tip when you start typing the name of the Function. You'll be able to make your selection from the list that appears. Don't, however, hit the Enter key! As an alternative, press the Tab key to continue inserting other options. You may end up with an error if you hit the Enter key. You can quickly fix this by selecting the cell again and completing your Function in the formula bar.

2. Using Insert Function Option from Formulas Tab

Using Excel's "Insert Function" dialog box, you can fully control the insertion of functions. To do this, go to the "Formulas" tab and then click on the "Insert function." The functions required for financial analysis are provided in the dialog box.

3. Selecting a Formula from One of the Groups in Formula Tab

This option is for users who want quick access to their favorite functions. To access this menu, head to the "Formulas tab" and then select your preferred group. Click on this tab to display a list of functions and then select your preferred option. Or, click on "More Functions" to access other options.

CHAPTER FIVE- MS EXCEL TEXT FUNCTION

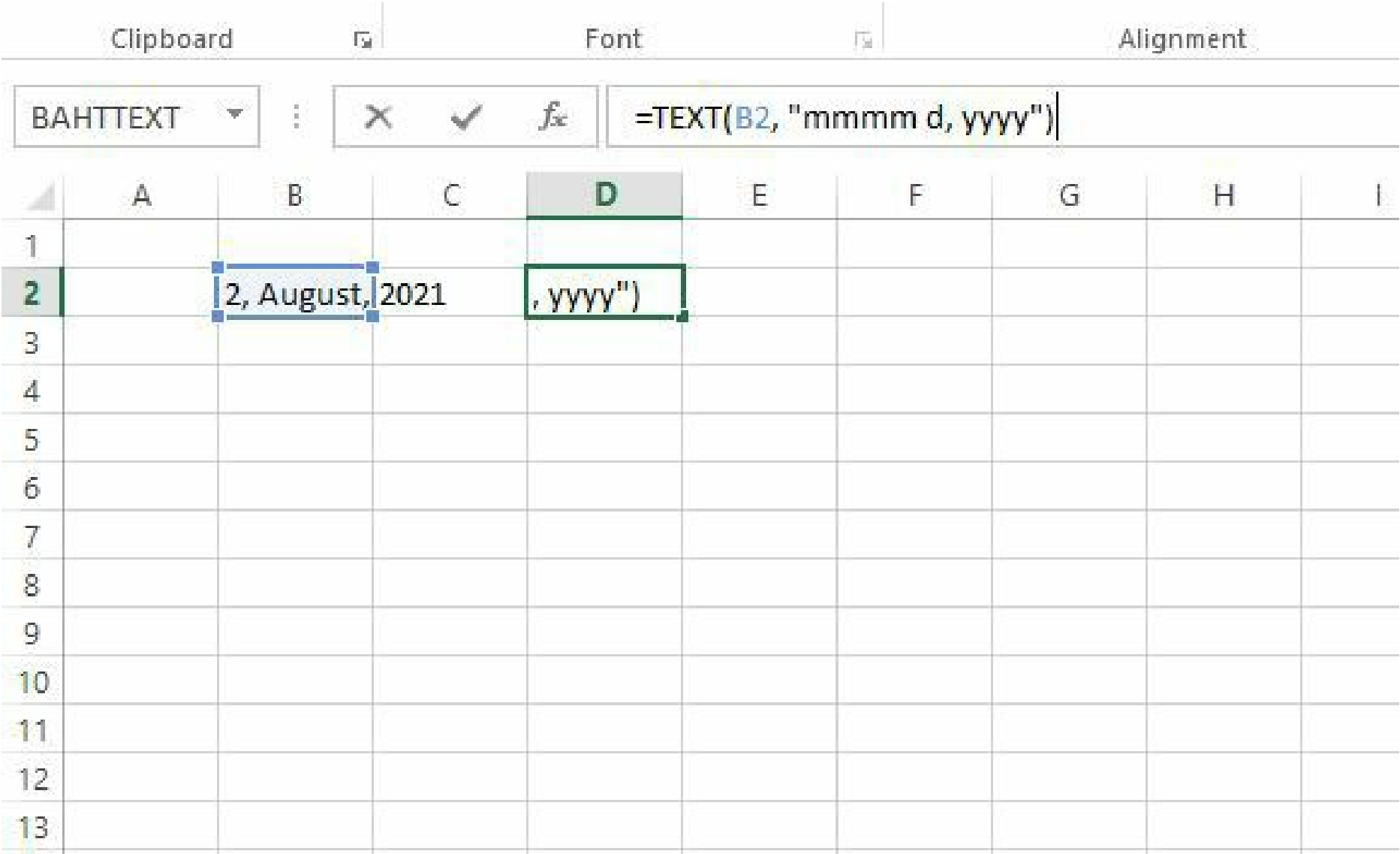
If you're working with worksheets, you'll undoubtedly have to deal with numbers. You may require the TEXT function in Excel at some point. TEXT is a function that converts a numeric number to a text string in a predefined format.

The syntax for this function is - TEXT(value, format_text)

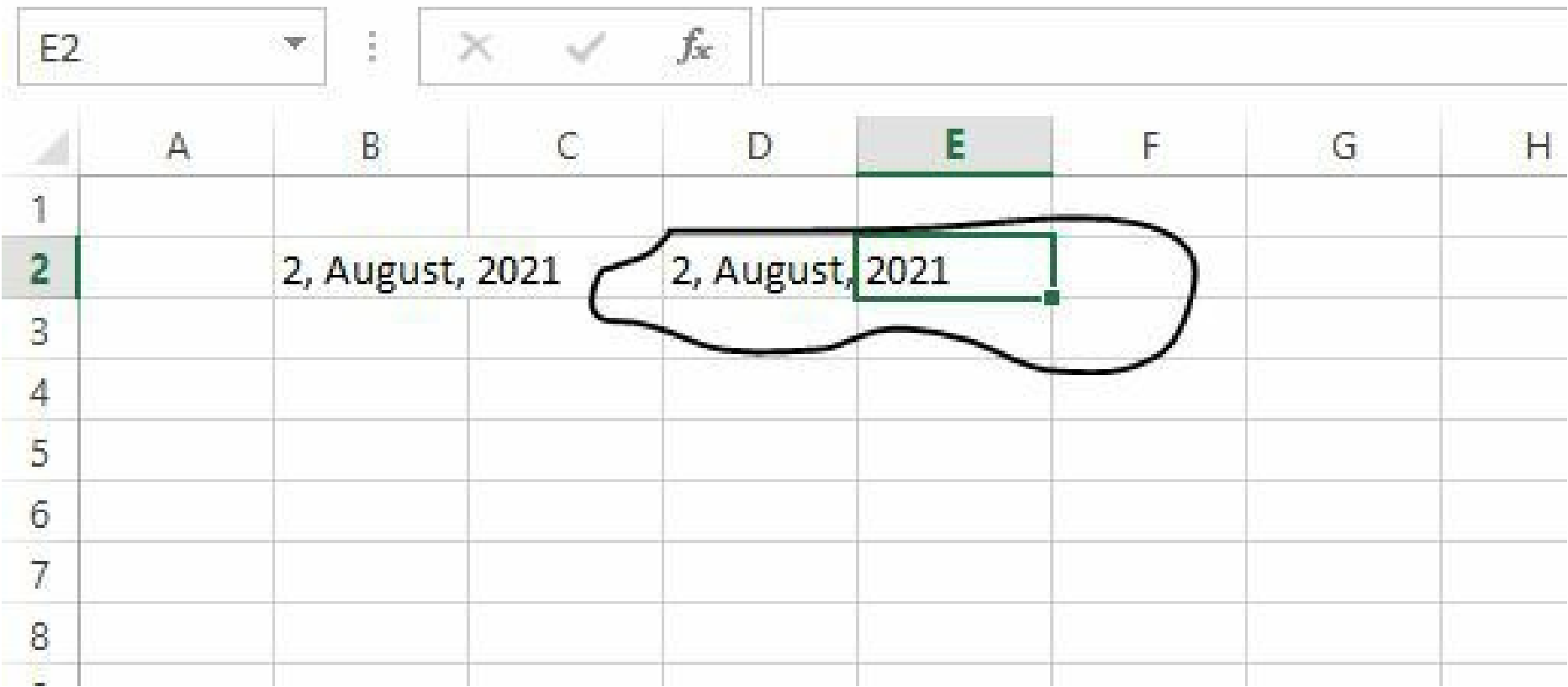
Where:

- Value – This is simply the numeric value to be converted to text.
- Format_text – this is the format you'd like to use. It comes in the form of a format code surrounded by quote marks. e.g. "dd/mm/yy".

For example, the formula in the screenshot below was used to retrieve the date "2 August 2021."



Insert the formula “=TEXT(B2, "m d, yyyy")” and then press the “Tab” key on your keyboard.



Note that the results of text formulas cannot be used in other calculations because the results are converted into strings. You can save the original values if you want them (just keep them out of sight) and then use them in other calculations.

Excel TEXT Function Format Codes

While this function has a simple syntax, as you've just seen, the most challenging aspect is figuring out how you're going to place your numbers the way you want. For the most part, the TEXT function takes Excel's numeric format codes. To help you understand better, here's a list of some commonly used codes.

Codes	Description	Format code example
#	Extra zeros will be ignored in this digit placeholder.	### up to two decimal places are displayed If you type 5.5457, it will be displayed as 5.55.
, (comma)	Thousands separator.	###,###.##. displays a thousand separator and 2 decimal places If you type 300000, it will be displayed as 300,000.00
?	Insignificant zeros are not displayed by a digit placeholder that leaves a space for them but does not display them. Commonly used to align decimal points in a column of numbers.	#.??- The decimal points in a column are aligned and displayed to a maximum of two decimal places.
0	Insignificant zeros are displayed in a digit placeholder.	#.00- 2 decimal places are always shown. If you type 4.6 in the referenced cell, it will be displayed as 4.60.

How to Use TEXT Function with Formula Examples

TEXT application in Excel may seem restricted because you can quickly change the presentation of numbers using the "Format Cells" dialog box, which is opened via the cell's context menu or the keyboard shortcut (Ctrl + 1). With the help of additional Excel functions, however, the situation is different. Hopefully, the following examples can help you come up with creative ideas.

Concatenate text and number (or date) in a custom format

When preparing summary sheets or reports, you may want to explain to your users what each statistic represents in addition to calculating totals, percentages, etc. Use the CONCATENATE function to connect text and numbers, and then use the TEXT function to display the number (or date) exactly how you want it.

Example 1. Format numbers inside a text string

For example, suppose you use the following formula to calculate the total based on the unit price in cell A3, the quantity in cell B3, and the discount percentage in cell C3: `= A3 * B3 * (1-C3)`. It should display a numerical result and some explanation. I suggest using easy words that won't make things complicated for you. You can use language like "your price." Additionally, it must include the currency sign, a comma, and two decimal.

You can do this by specifying the above calculation in the first argument of the TEXT function, adding suitable formatting codes in the second parameter, and using the CONCATENATE function to concatenate the text formula with the string,

=> "Your price is "&TEXT(A3*B3*(1-C3), "\$###,###.00")

Or

= CONCATENATE("Your price is ",TEXT(A3*B3*(1-C3), "\$###,###.00"))

Take a look at the screenshot below

The screenshot shows an Excel spreadsheet with the following data:

	A	B	C
1			
2	Price	qty	Percent-off
3	\$20	5	30%
4			
5	"Your price is \$70.00")		

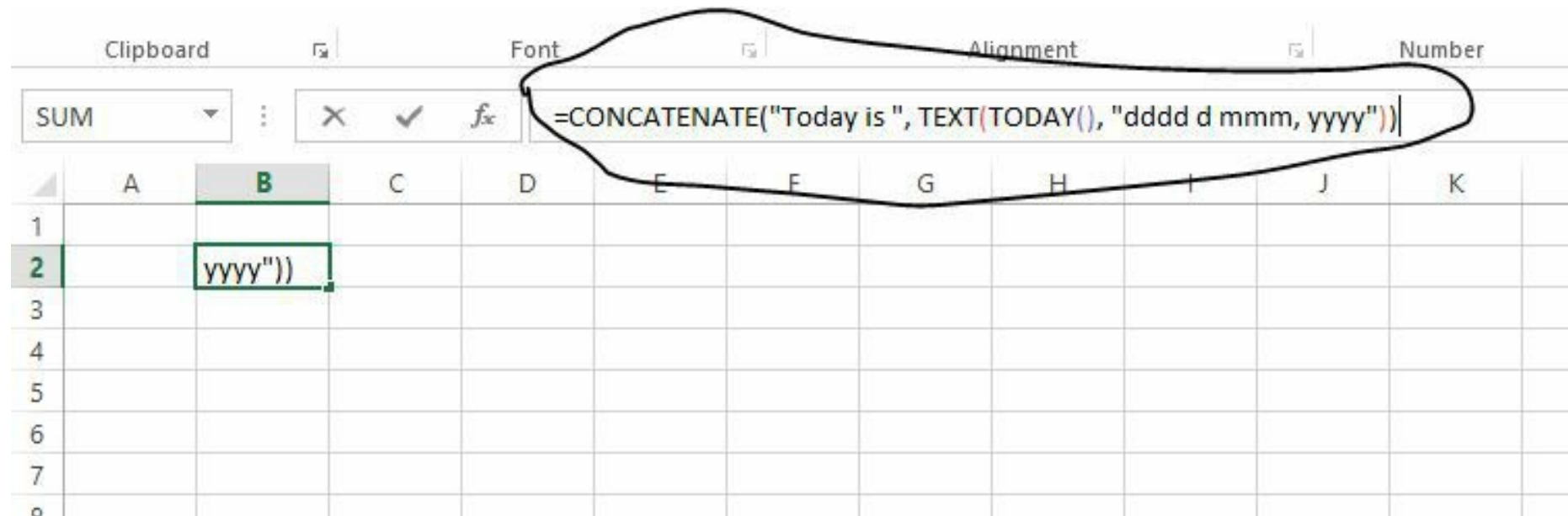
The formula bar shows the formula being entered in cell B5: `= "Your price is "&TEXT(A3*B3*(1-C3), "$###,###.00")`.

Example 2. Combine text and date in the desired format

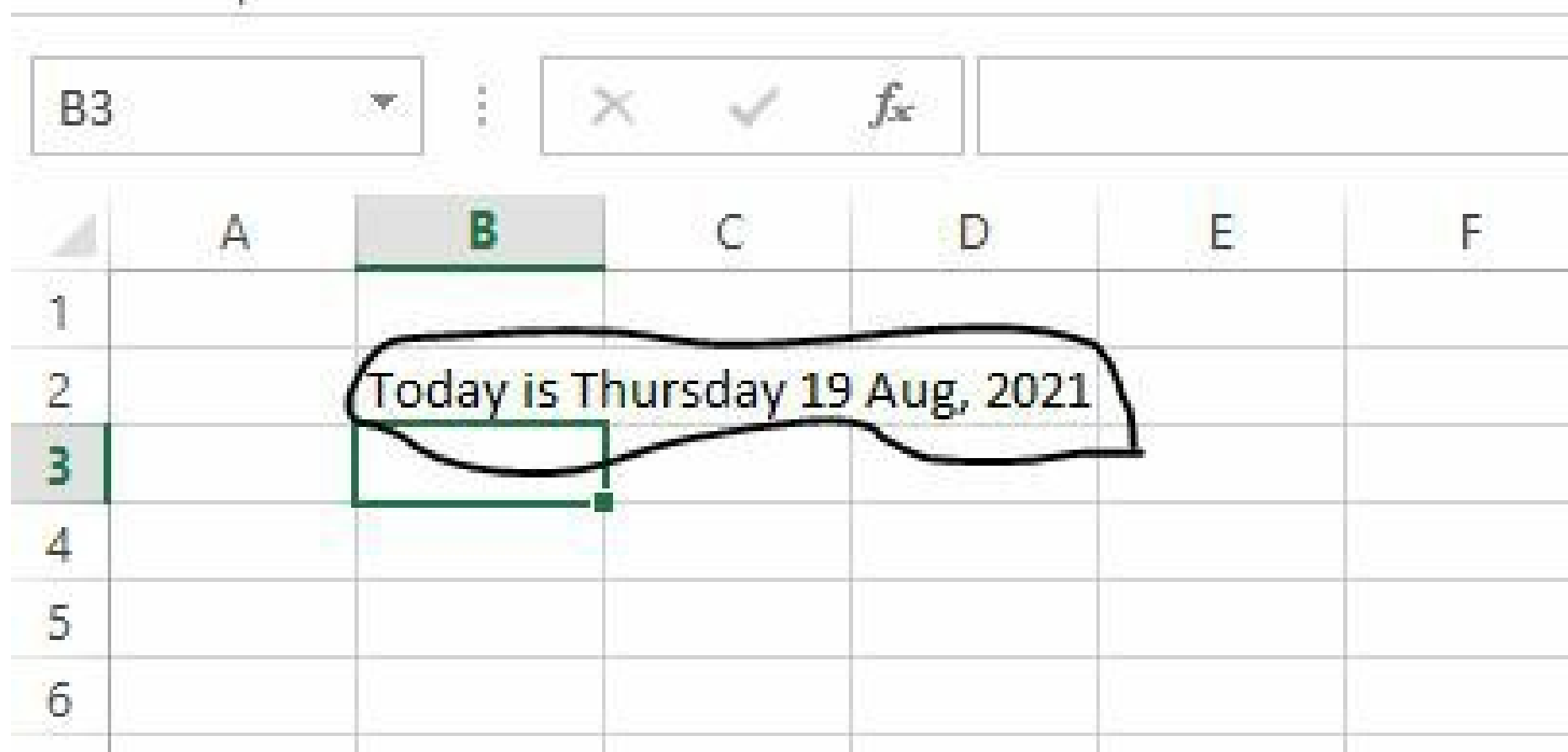
When using the TODAY () or NOW () function to return the current date, you may want to write it with some text so that no one is confused about which day the date belongs. If you concatenate a text and a date the usual way-
`=CONCATENATE("Today is ", TODAY())`, you may end with an error like "Today is 31217".

For example, using the TODAY function in the Text formula with the format code "dddd d mmm, yyyy" returns a string that looks like this: "Today is Thursday 19 Aug, 2021." The following is the complete formula:

`=CONCATENATE("Today is ", TEXT(TODAY(), "dddd d mmm, yyyy"))`



Then press the Enter or TAB key



3. Add leading zeros to numbers with variable lengths

You may already know that Microsoft Excel deletes leading zeros before numbers in cells. This is fine in most cases, but what if you want to keep preceding zeros? For example, to display 6-digit numbers with leading zeros, use the following formula (where C2 is the original number):

`=TEXT(C2,"0000000")`

As seen in the screenshot, our text formula adds as many leading zeros as are required to produce a 6-digit string (remember, the result of the TEXT function is always a text string, even if it Looks like a number).

✕ ✓ <i>fx</i> =TEXT(C2,"00000000")		✕ ✓ <i>fx</i>			
	C	D	E	F	G
	First value	Formatted value			
	14	(C2,"00000000")			

Compared to other Excel functions, the TEXT function is easy to grasp, and you'll have no trouble incorporating it into your spreadsheets. If a Text formula fails to work, it's almost often due to entering an improper format code in the format text argument.

Also, note that the formula depends on your language/region. For example, this formula, TEXT(A2, "mm/dd/yy"), will work perfectly for English users, but German users may have an invalid value. You will need to change the format codes to "t" instead of "d" and "j" instead of "y" because "day" in German is "tag" and year is "Jahr." m" is fine because the German word for month begins with an "m" (Monat). So, in German Excel, the above formula will read as follows: =TEXT(A2; "mm/tt/jj")

Also, keep in mind that the List Separator in European countries is usually set to semicolon rather than a comma. Therefore ";" should be used to separate the formula's arguments.

Other Excel Text Functions

In addition to TEXT, Excel also provides several other functions you can use to process text values. Some are used to convert text to numbers and vice versa, while others are used to manipulate text strings directly.

Excel LEFT function

The Left function is a commonly used Text Function that allows you to extract specific characters from the left side of a text string. In Excel, the LEFT function returns the required number of characters (substrings) from the beginning of a string. The syntax of the LEFT function is as follows:

LEFT(text, [num_chars])

Where:

- The text is the text string from which you want to extract the substring. This usually refers to a cell with text in it.
- Num_chars (optional) – this is the number of characters to extract.
 - If num_chars is left blank, it defaults to 1, which means a Left formula will only yield one character.
 - A Left formula will return the entire text if the num_chars is larger than the overall length of the text.

For example, the syntax “=LEFT(B2, 4)” will extract the first four characters from the text in cell B2. The screenshot below shows the result:

The screenshot shows the Excel interface. The formula bar at the top displays `=LEFT(B2, 4)`. Below it, a worksheet is visible with columns A, B, C, D, and E. Row 1 contains the headers 'Original string' in B1 and 'first 3 chars' in C1. Row 2 contains the text 'They love hiking' in B2 and the result of the formula, 'They', in C2. The formula `=LEFT(B2, 4)` is entered in cell C2. The result 'They' is displayed in cell C2.

	A	B	C	D	E
1		Original string	first 3 chars		
2		They love hiking	They		
3					
4					
5					
6					
7					

Use the LEFT function in conjunction with the VALUE function if you are dealing with a numeric dataset and want the LEFT function to return a number.

1. Using the Left function to extract a substring before a specific character

The text string preceding a specific character may be of interest in some instances. It's possible, for example, to retrieve only the first names or country codes in a cell. The only problem is that each name and code has a different number of characters, so you can't just put a predefined value in the num_chars argument of your Left formula like we did in the previous example.

If a space separates the first and last names, the problem is solved by specifying the position of the space character in the string, which can be done quickly using the SEARCH or FIND functions.

Use the following formula- `LEFT(B2, SEARCH(" ", B2))` to do this.

Note that trailing spaces can cause a lot of problems, so get rid of them by subtracting 1 from the Search formula result (not visible in cells)

`=LEFT(B2, SEARCH(" ", B2)-1)`

fx				
=LEFT(B2, SEARCH(" ", B2)-1)				
	A	B	C	D
1		Full Name	First Name	
2		Jessica Peters	(" ", B2)-1)	
3		Ian Glen		
4		Alfred Alfie		
5				
6				
7				

A	B	C	D	E
	Full Name	First Name		
	Jessica Peters	Jessica		
	Ian Glen			
	Alfred Alfie			

Finally, this generic formula can be used to create a substring that comes before any other character:
LEFT(string, SEARCH(character, string) - 1)

2. How to remove the last N characters from a string

Finally, this generic formula can be used to create a substring that comes before any other character:

LEFT(string, SEARCH(character, string) - 1)

2. How to remove the last N characters from a string

To do so, combine the LEFT and LEN functions like this: LEFT(string, LEN(string) - number_of_chars_to_remove)

The formula works like this: the LEN function gets the total number of characters in a string, then you subtract the number of characters you don't need from the total length, and the LEFT function returns the remaining characters.

For example, to remove the last 5 characters from a text in B2, use this formula:

=LEFT(B2, LEN(B2)-5)

This example is demonstrated in the screenshot below.

=LEFT(B2, LEN(B2)-5)

	A	B	C	D	E
1		Source string	Result		
2		Invalid:223 - Do	=LEFT(B2, LEN(B2)-5)		
3		Ian Glen			
4		Alfred Alfie			
5					
6					
7					
8					
9					
10					

	A	B	C	D
1		Source string	Result	
2		Invalid:223 - Do	Invalid:223	
3		Ian Glen		
4		Alfred Alfie		
5				
6				
7				
8				
9				

3. How to force the LEFT function to return a number

As you know, even if you only want the first few digits of a number, the LEFT function always outputs text. Because of this, you will not be able to use the results of the formulas on the left in calculations or other Excel functions that work with numbers.

So, how do we make the LEFT function produce a number instead of a string of text? Wrap it in the VALUE function, which converts a string expressing a number to a number, as shown below:

VALUE(LEFT())

Use the following formula to extract the first two characters from the string in B2 and convert the result to numbers:
=VALUE(LEFT(B2,2))

Your result will look like this:

Font		Alignment	
<i>f_x</i>	=VALUE(LEFT(B2,2))		
	A	B	C
1		Source string	First 2 chars
2		21 -23	21
3		22 -40	22
4		33 -45	33
5			
6			
7			
8			
9			

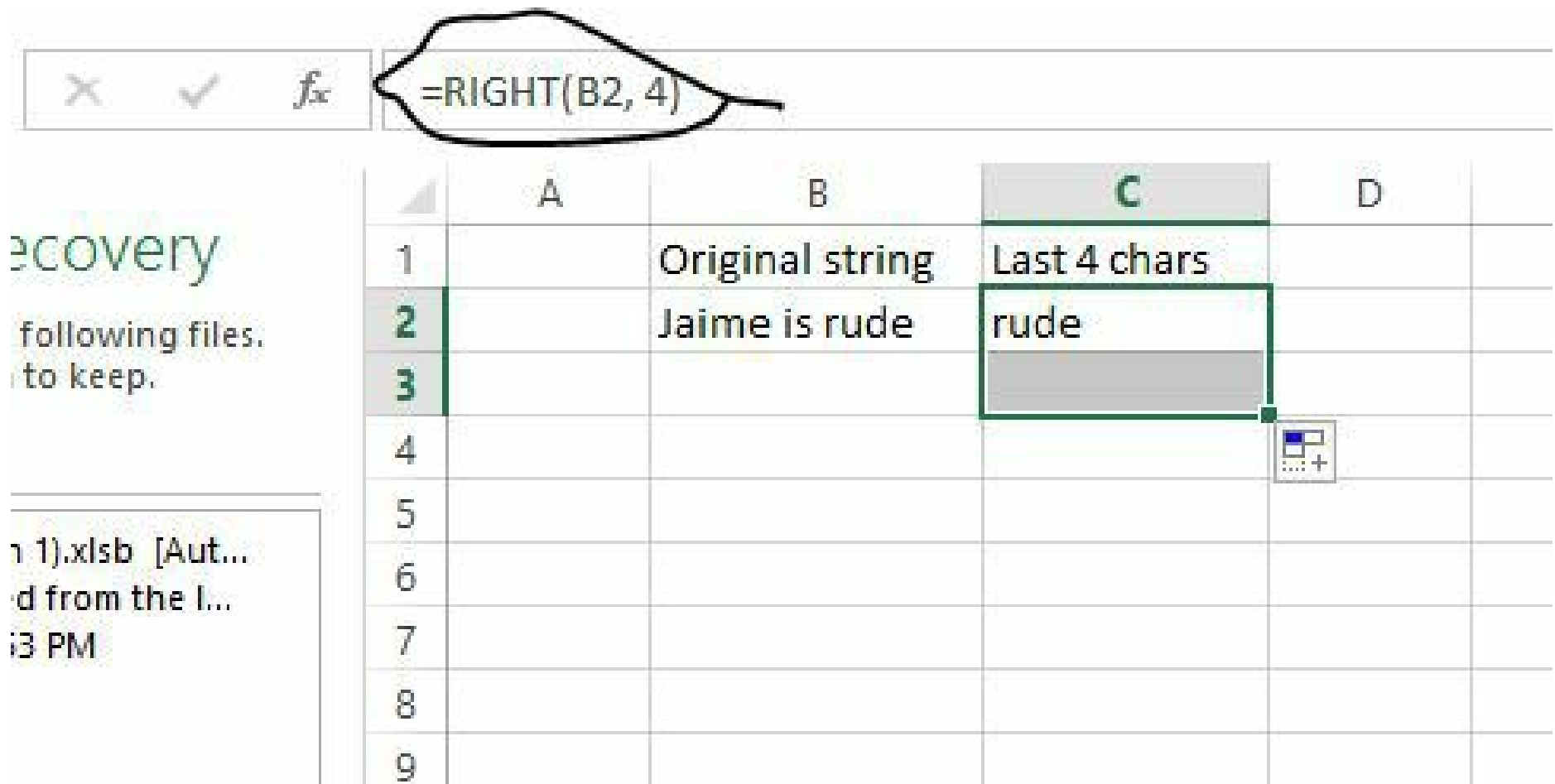
RIGHT function

Excel's RIGHT function returns the specified number of characters from the end of a text string. The RIGHT function has the following syntax:

RIGHT(text, [num_chars])

- When num_chars is not specified, the string's last character is returned (default).
- All characters are returned if num_chars exceeds the total number of characters in the string.
- If num_chars is a negative number, a Right formula returns the #VALUE! Error.

For example, the following formula- "`=RIGHT(B2, 4)`" is used to extract the last 4 characters from the string in cell B2. The result will look like this:



	A	B	C	D
1		Original string	Last 4 chars	
2		Jaime is rude	rude	
3				
4				
5				
6				
7				
8				
9				

Note: Even if the initial value of the RIGHT function is a number, it always returns a text string. Combine the VALUE function with the Right formula to make it generate a number.

1. How to use the RIGHT function to get a substring that comes after a specific character

A substring can be extracted using the SEARCH or FIND functions to find a character's position, subtracting it from the total length of the original string supplied by the LEN function, and pulling that many characters off its rightmost end.

`RIGHT(string, LEN(string) - SEARCH(character, string))`

Let's assume cell B2 contains the first and last name separated by a space, and you'd like to move the last name to another cell in the spreadsheet. Take the generic formula above and substitute B2 for string and " " (space) for a character, and you get:

`=RIGHT(B2,LEN(B2)-SEARCH(" ",B2))`

✕

✓

fx

=RIGHT(B2,LEN(B2)-SEARCH(" ",B2))

recovery

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3 PM

	A	B	C	D
2		Jessica Peters	Peters	
3		Ian Glen	Glen	
4		Alfred Alfie	Alfie	
5				
6				
7				
8				
9				
10				
11				

To retrieve a substring following any other character, use the following formula =RIGHT(B2,LEN(B2)-SEARCH("-",B2)) to extract the substring that comes after a hyphen.

✕

✓

fx

=RIGHT(B2,LEN(B2)-SEARCH("-",B2))

recovery

following files.
to keep.

1).xlsx [Aut...

d from the l...

3 PM

	A	B	C	D
2		Jessica-Peters	Peters	
3		Ian-Glen	Glen	
4		Alfred-Alfie	Alfie	
5				
6				
7				
8				
9				
10				

TRIM function - quick way to remove extra spaces

Are your algorithms behaving weirdly? Or do you keep getting just zeros when you add two columns of numbers? And why is your Vlookup formula, which appears correct, throwing so many N / A errors? All these issues and more are all caused by unnecessary spaces in your cells

1. TRIM function - remove extra spaces

The TRIM function removes extra spaces from a text. It removes all leading, trailing, and middle spaces except for one space between words. The syntax for the TRIM function- “TRIM(text)” is the easiest one could imagine.

For example, use the following formula =TRIM(B2) to remove spaces in cell B2. Your cell should look like this:

Font

Alignment

Number

✓ *f_x*

=TRIM(B2)

ery

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	A	B	C	D
1		Original Data	Trimmed Spaces	
2		Jessica Peters	Jessica Peters	
3		Mark Lucas	Mark Lucas	
4		Jonny Glover	Jonny Glover	
5				
6				
7				
8				
9				
10				

That’s it!

A combination of the TRIM function and CLEAN can be used to remove line breaks and non-printing characters. For example, use this formula- =TRIM(CLEAN(B2)) to remove spaces, line breaks, and other unwanted characters from cell B2

2. How to remove leading spaces in a numeric column

As you can see, the TRIM function in Excel has successfully removed all extra spaces from the text data column, but what if your data is numeric? At first glance, the TRIM function seems to have done its job. If you take a closer look, you will find that trimmed values do not behave like numbers. Here are some examples of abnormalities:

- Even if you apply numeric formatting to cells, the original column with leading spaces and trimmed numbers will be left-aligned, and normal numbers will be right-aligned by default.
- Excel shows COUNT on the status bar only if two or more cells with trimmed numbers are selected. It will also show SUM and AVERAGE for numbers.
- When the SUM formula is applied to the trimmed cells, the result is zero.

When searching for numbers, trimmed values appear as text strings. To fix this, multiply the trimmed values by one (use the Paste Special> Multiple options to multiply all the numbers at once).

Enclosing the TRIM function in VALUE, as shown below, is a better option.

=VALUE(TRIM(B2))

This formula will remove any leading and trailing spaces, turning the resulting value into a number. Your cell should look like this:

C2

:

✖

✓

fx

=VALUE(TRIM(B2))

	A	B	C	D	E	F
1		Original data	Trimmed numbers			
2		1111	1111			
3		333333	333333			
4		2233	2233			
5						
6						
7						
8						

3. How to remove only leading spaces (Left Trim)

We'll use a slightly more complex formula to keep all the spaces in between:

=MID(B2,FIND(MID(TRIM(B2),1,1),B2),LEN(B2))

To find the first text character in any, string using this formula requires the use of FIND, MID, and TRIM together. Your cell should look like this:

Font		Alignment		
=MID(B2,FIND(MID(TRIM(B2),1,1),B2),LEN(B2))				
B	C	D	E	F
Area codes	Trimmed values			
232, Seattle,	232, Seattle,			
123, Texas, Mb	123, Texas, Mb			
111, York, RD	111, York, RD			
212, Castle, town	212, Castle, town			

Tip. The "Trim Spaces" tool can also be used to remove spaces from the ends of cells. There isn't a simple Excel formula for deleting leading and trailing spaces while maintaining a large number of gaps between words simultaneously.

CHAPTER SIX- MS EXCEL LOGIC FUNCTION

Four logical functions are available in Excel to work with the logical values- AND, OR, XOR and NOT. Logic functions in Excel return TRUE or FALSE when their arguments are evaluated, just like logical operators do. A summary of each logical function is provided in the following table to assist you in selecting the right formula for your tasks.

Function	Description	Formula Example	Formula Description
NOT	Returns the argument's logical value in reverse. To put it another way, if the argument is FALSE, TRUE is returned, and vice versa.	=NOT(A2>=10)	If a value in cell A1 is more than or equal to ten, the formula yields FALSE; otherwise, it returns TRUE.
OR	If any argument evaluates to TRUE, this function returns TRUE.	=OR(A2>=10, B2<5)	If A2 is higher than or equal to 10 and B2 is less than 5, or both conditions are met, the formula returns TRUE. The formula returns FALSE if neither of the conditions are met.
XOR	All arguments are combined into a logical Exclusive Or.	=XOR(A2>=10, B2<5)	If A2 is higher than or equal to 10 and B2 is less than 5, the formula returns TRUE. The formula returns FALSE if neither of the conditions is met or both conditions are met.
AND	Checks a number of conditions and returns true if all of them evaluate to true.	=AND(B2>=10, C2<5)	The formula returns TRUE if the value in cell B2 is greater than or equal to 10 and the value in cell C2 is less than 5, otherwise FALSE.

Microsoft Excel also has three other "conditional" functions: IF, IFERROR, and IFNA, in addition to the four logical functions listed above.

Using the AND Function

The AND function is very useful for testing multiple conditions and making sure they are all met. Technically the AND function tests the specified condition and returns TRUE if all conditions evaluate to TRUE. Otherwise, it returns FALSE. The syntax for the AND function in Excel is:

AND(logical1, [logical2], ...)

Where logical represents the condition you want to test, which can be TRUE or FALSE.

The first (logical1) condition is required; the other conditions are optional. Now let’s take a deeper dive into this function.

Formula	Description
=AND(A2="Laptop", B2>C2)	If A2 contains "Laptop" and B2 is greater than C2, the result will be TRUE; otherwise, FALSE.
=AND(B2>21, B2=C2)	If B2 is more than 21 and B2 equals C2, the result will be TRUE; else, FALSE.
=AND(A2="Laptop", B2>=32, B2>C2)	If A2 contains the word "Laptop," B2 is greater than or equal to 32, and B2 is greater than C2, the result is TRUE; otherwise, the result is FALSE.

Now, use the screenshot below to create a similar worksheet and then input the formulas listed in the table above. Your results will return TRUE or FALSE

D3

	A	B	C	D	E	F	G
1	Item	Qty	Sold	Formula 1	Formuls 2	Formula 3	
2							
3	Laptop						
4	Phone						
5	Mouse						
6							
7							
8							

Excel AND function - common uses

The AND function alone is not fascinating, and its application is limited. In addition, you can combine the AND function with other Excel functions, such as the IF function. For example, you can add one of the AND functions above inside the IF function and get a result similar to this:

=IF(AND(A2="Laptop", B2>C2), "Real", "Fake")

[illegible]

Using the OR Function In Excel

As with the AND function, Excel's OR function compares two values or statements; however, the OR function returns TRUE when one of the arguments is TRUE and FALSE when all arguments are false. The syntax for this function is:

OR(logical1, [logical2], ...)

As you can see, it is very similar to the AND function. Now let's look at some examples.

Formula	Description
=OR(A2="Keyboard", A2="Laptop")	If A2 contains "Keyboard" or "Laptop," it returns TRUE; otherwise, it returns FALSE.
=OR(B2>=32, C2>=18)	Returns TRUE if B2 is more than or equal to 32 and FALSE if B2 is less than or equal to 18.
=OR(B2=" ", C2="")	Returns true if B2 or C2 or both are blank; otherwise, returns FALSE.

Create a worksheet similar to the one shown below, and then enter the formulas from the table above. Your answers will either be TRUE or FALSE.

	A	B	C	D	E	F	G
1	Item	Qty	Sold	Formula 1	Formula 2	Formula 3	
2							
3	Keyboard	20	19				
4	Laptop	40	26				
5	Phone	70	66				
6	Mouse	19	6				
7							
8							
9							
10							
11							
12							

IF function with nested OR

Here’s an example of the OR function combined with the IF function

=IF(OR(B2>30, C2>20), "Real", "Fake")

Clipboard		Font		Alignment				
D2	:				=IF(OR(B2>30, C2>20), "Real", "Fake")			
	A	B	C	D	E	F	G	H
1	Item	Qty	Sold	IF formula				
2	Keyboard	20	19	Fake				
3	Laptop	40	26	Real				
4	Phone	70	66	Real				
5	Mouse	19	6	Fake				
6								
7								
8								
9								

Using the XOR Function

The XOR function is a logical Exclusive OR function. The XOR function has the same syntax as the OR function- XOR(logical1, [logical2],...)

The first logical statement (Logical 1) is required, and the rest are optional. You can test up to 200 conditions in a single formula that are evaluated to either TRUE or FALSE. The formulas below will give you a better understanding of this function:

Formula	Result	Description
=XOR(3>1, 5<2)	TRUE	Because the first argument is TRUE and the second argument is FALSE, the result is TRUE.
=XOR(3<1, 5<2)	FALSE	Returns FALSE because both arguments are FALSE.
=XOR(3>1, 5>2)	FALSE	Because both arguments are TRUE, the result is FALSE.

In case you are still confused, consider the following example. Assuming you have a table of Students and their results for the last two grades. You want to know which student will move to the next class on the following conditions:

- Students who passed Grade 1 and 2 advance to the next class automatically and won't have to repeat any class.
- Students who failed both Grade 1 and 2 will leave the school
- Students who passed only Grade 1 or 2 will have to sit for another exam to determine if they move to the next class.

A simple XOR formula- =XOR(B2="Passed", C2="Passed") will give you the following result:

D2	:				=XOR(B2="Passed", C2="Passed")				
	A	B	C	D	E	F	G	H	
1	Student	Grade 1	Grade 2	Repeat					
2	Jessica	Passed	Passed	FALSE					
3	James	Passed	Failed	TRUE					
4	Jaime	Failed	Passed	TRUE					
5	Jeremy	Failed	Failed	FALSE					
6									
7									

You can get better results by nesting this XOR function into the IF formula:

=IF(XOR(B2="Passed", C2="Passed"), "Yes", "No")

Using the NOT Function In Excel

This function has a straightforward syntax- NOT(logical)

This function reverses the value of its argument. In other words, the NOT function returns FALSE if an argument is TRUE and vice versa. For example, both formulas below return TRUE:

=NOT(FALSE)

=NOT(3*3=4)

So you might ask, why would anyone want such weird results? In some cases, knowing when a condition fails is more important than knowing when a condition occurs. For example, let's say you want to leave out Acer laptops from a list of laptops you made. I hope this example helps: =NOT(C2="Acer")

E2					=NOT(C2="Acer")	
	A	B	C	D	E	F
1	Item	Color	Model	Price	Any model but Acer	
2	1112	White	HP	\$600	TRUE	
3	1113	Black	Acer	\$500	FALSE	
4	1114	Gold	Dell	\$600	TRUE	
5	1115	Black	Acer	\$600	FALSE	
6	1116	Black	Dell	\$500	TRUE	
7						
8						
9						

As you may know, there are several ways to do things in Excel. To test several conditions in a single formula, use the NOT function in conjunction with the AND or OR function. For example, if you want to exclude Acer and Dell models, the formula would go like:

=NOT(OR(C2="acer", C2="dell"))

And if you'd rather not have an Acer laptop, while an Acer charger may be considered, then combine the NOT function with the AND function:

=NOT(AND(C2="acer", B2="charger"))

The NOT function is often used to reverse the behavior of another function. For example, you can create the ISNOTBLANK formula by combining the NOT and ISBLANK functions.

As you know, the =ISBLANK() will return TRUE if a cell is blank. The NOT function will reverse the result to FALSE: =NOT(ISBLANK(B1))

Then you can go ahead and create nested IF statements using NOT / ISBLANK functions for your tasks:

=IF(NOT(ISBLANK(C2)), C2*0.15, "No profit :(")

D6				=IF(NOT(ISBLANK(C6)), C6*0.15, "No profit :(")					
	A	B	C	D	E	F	G	H	I
1	Item	Purchase Price	Sale Price	Profit					
2	Headsets	\$400	\$500	75					
3	Airpods	\$600	\$700	105					
4	Chargers	\$800		No profit					
5	Phones	\$2,000		No profit					
6	Laptops	\$3,000	\$3,200	480					
7									
8									
9									

In plain English, the formula instructs Excel to perform the following actions: For each additional sale, multiply the number in C2 by 0.15 if the field C2 is not empty. Whenever C2 is left blank, the message "No profit will be displayed on the screen.

CHAPTER SEVEN- FUNCTIONS OF DATE AND TIME

Using this function is as easy as it sounds. You can address complex issues by merging several functions into one formula.

Date functions	Description
DATE	This is the main function
TODAY	Returns the current date
NOW	Returns
DATEVALUE	Converts a text date into a date format
TEXT	Converts a date to a text value
DAY	Returns the day of the month
MONTH	returns the month of a specified date
EDATE	Returns a date N months before or after the start date
DATEDIF	Returns the difference between two dates
YEARFRAC	Calculates the fraction of the year between 2 dates

Excel DATE Function

DATE(year, month, day) returns a date's serial number depending on the values you provide.

If you're going to work with dates in Excel, the DATE function is a must-know. Because other Excel date functions may not always recognize dates entered in text format, this is an important factor. You should use the DATE function to supply dates when performing date calculations in Excel to ensure accurate results.

The Excel DATE function appears easy at first glance, but this tutorial points out several specificities. Here are a few DATE formula examples:

=DATE(2021, 8, 20) - returns a serial number corresponding to 20-August-2021.

=DATE(YEAR(TODAY()), MONTH(TODAY()), 1) - returns the first day of the current year and month.

=DATE(2021, 8, 21)+5 - adds 5 days from August 20, 2021.

Excel TODAY function

The TODAY() function returns today's date. The following formula can be entered in a cell whenever you need to know the current date.

=TODAY()

It's also possible to use this function in more complex formulas and calculations. Enter the following formula in a cell to subtract 5 days to the current date:

=TODAY()-7

Use this formula to add 20 weekdays to today's date, excluding weekends:

=WORKDAY(TODAY(), 20)

Shortcuts to get today's date

Ctrl + ; to get today's date

Ctrl + Shift + ; to get the current time

To enter the current date and time: Ctrl + ; then Space and then Ctrl + Shift + ;

Excel NOW Function

The NOW() function returns the current date and time. It has no counter-arguments, just like TODAY. The following formula- =NOW() displays the current date and time in your worksheet.

This function refreshes the returned value each time a worksheet is recalculated, as does TODAY. Press Shift+F9 to force the active worksheet to perform a recalculation, which will update your NOW formula's value.

Excel DATEVALUE function

In this case, DATEVALUE(date_text) will convert a date to a number that represents a date.

There are several date formats that the DATEVALUE function can handle. You can use this function to convert "text dates" into the Date format. Listed below are a few examples of simple DATEVALUE formulas.

=DATEVALUE("20-August-2021")

=DATEVALUE("8/21/2021")

=DATEVALUE("august 20, 2021")

Excel DAY function

Day(serial number) returns an integer from 1 to 31, representing a day of the month. The day you are trying to retrieve is represented by a serial number. Cell references, dates provided with the DATE function, and formula returns can all be used. Here are a few examples:

=DAY(B2) - returns the day of the date in B2

=DAY(DATE(2021,8,23)) - returns the day of 23-August-2021

=DAY(TODAY()) - returns the day of today's date

This same format can also be used for the MONTH and YEAR function. E.g.

=MONTH(B2)

=MONTH(TODAY())

=YEAR(B2)

=YEAR(TODAY())

Excel WEEKDAY Function

A date is sent to the WEEKDAY(serial number,[return type]) function, which produces a number from Sunday through Saturday. Date, cell reference, or date returned by another function can all be used as Serial_number.

Return_type (optional) specifies which day of the week should be treated as the first day of the week.

As an example, have a look at this WEEKEND formulas.

=WEEKDAY(B2) - This function returns a Sunday since the first day of the week corresponds to a date in cell B2 (Sunday) (default).

=WEEKDAY(B2, 2) - The day of the week that corresponds to the date in cell B2 is returned; the week starts on Monday.

=WEEKDAY(TODAY()) - This function returns the day of the week for today. The week begins on Sunday.

The WEEKDAY function can help you differentiate workdays and weekdays and sort, filter, and highlight them.

Excel DATEDIF function

The DATEDIF(start date, end date, unit) function is used to determine the difference between two dates. When computing the date difference, the period to use depends on the letter you supply as the last argument:

=DATEDIF(B2, TODAY(), "d") - calculates the difference in days between B2 and today's date.

=DATEDIF(B2, B5, "m") - The number of complete months between B2 and C2 is returned.

=DATEDIF(B2, B5, "y") - The number of complete years between B2 and C2 is returned.

CHAPTER EIGHT- COUNTING FUNCTION

This function counts cells, as the name suggests. But wait, there's more to it! To perform different counting tasks, Excel has five different COUNT types:

- COUNT
- COUNTA
- COUNTBLANK
- COUNTIF
- COUNTIFS

I am only going to treat a few of these functions.

COUNT function

The COUNT function accepts comma-separated values as inputs. In this case, the function will return the total number of the numeric values specified. The syntax is as follows:

=COUNT(value1,[value2],...)

Usage:

B2		:	✕ ✓ <i>fx</i>			=COUNT(100, 200, 300)	
	A	B	C	D	E	F	G
1							
2		3					
3							
4							
5							

The COUNT function is used to find the total number of values in a parameter that contains multiple values.

COUNT(range)

This function counts the number of cells in a range. As you can see, the number of cells in range A2:A6 can be determined using =COUNT(A2:A6). To use this function, you must supply only numeric values.

ClipboardFont

E2					=COUNT(A2:A6)	
	A	B	C	D	E	F
1	S/N	Student	Age	Gender		
2	1	Jessie Jane	18	Female	5	
3	2	John Stone	18	Male	4	
4	3	Vanessa Steel	17	Female	3	
5	4	Bruce Wills	19	Male	2	
6	5	Jane Chastein	20	Female	1	
7						
8						
9						

The COUNTA function

Cells that contain data, i.e., non-empty values, are counted using this function. The syntax is as follows:

=COUNTA(value1,[value2],...)

E2	:	X	✓	<i>fx</i>	=COUNTA(C2:C6)
	A	B	C	D	E
1	S/N	Student	Age	Gender	
2	1	Jessie Jane	18	Female	3
3	2	John Stone	18	Male	2
4	3	Vanessa Steel		Female	1
5	4	Bruce Wills		Male	1
6	5	Jane Chastein	20	Female	1
7					
8					
9					

The values can be in any range. For example

As shown in the screenshot above, since C2 to C6 are all non-empty cells, =COUNTA(C2:C6) counts the number of cells with data.

How Do I Use COUNTIF?

The COUNTIF function counts the number of cells that meet a specific requirement. Here's how the syntax is written- =COUNTIF(range,criteria)

- Range- where a range of cells can be given. To give an example, let's say cell A2:A6
- Criteria- is a condition on the function that must be met. For example: ">50", A1

Usage:

You can use the COUNTIF function to check whether a string matches the criterion.

For example, If you want to find out how many times the name "John" appears, use the formula =COUNTIF(B2:B6, "John").

You can also use the COUNTIF to count cells that contain text-

Counting text-only cells are easy. We use "*" as our condition.

The asterisk or '*' symbol can be used to find any number of characters. When using '*' as a wildcard, you can match any number of characters.

In the range, D2 to E12, the function =COUNTIF(D2:E6, "*") counts the number of text-filled cells. Wildcard characters are helpful when it comes to counting cells that contain a specific word or letters. Take a look at the following example.

F4		:	X	✓	<i>fx</i>	=COUNTIF(D2:E6,"*")	
	A	B	C	D	E	F	G
1	S/N	Student	Age	Gender	Location		
2	1	Jessica	18	Female	China		
3	2	John	18	Male	London		
4	3	Bruce		Female	Brazil	10	
5	4	John		Male	Brazil		
6	5	Vanessa	20	Female	France		

The = COUNTIF (D2: E6, "*") function runs in the range D2 to E6 and counts cells that contain text.

Here are a few examples to help you understand better

Logical operation	Formula example	Description
Less than “<”	=COUNTIF(B1:B10,”<10”)	Count the number of values in a cell that are lesser than 20
Less than or equal to “<=”	=COUNTIF(B1:B10,”<=20”)	Count the number of values in a cell that are lesser than or equal to 20
Greater than “>”	=COUNTIF(B1:B10,”>20”)	Count the number of value in each cell that are greater than 20
Greater than or equal to “>=”	=COUNTIF(B1:B10,”>=20”)	Count the number of values for each cell that are greater than or equal to 20
Equal to “=”	=COUNTIF(B1:B10,”=20”)	Count if the values in cells are equal to 20
Not equal to “!=”	=COUNTIF(B1:B10,”<>20”)	Count the number of values for each cell that are not equal to 20

With the basics of these formulas, you can now expand your knowledge by addressing real-life tasks and creating smart complex formulae for your worksheets